

Report on the investigation of
the collision between the passenger vessel
Millennium Time
and the motor tug
Redoubt
with 3 barges in tow
on the Kings Reach, River Thames, London
17 July 2014



Extract from
The United Kingdom Merchant Shipping
(Accident Reporting and Investigation)
Regulations 2012 – Regulation 5:

“The sole objective of the investigation of an accident under the Merchant Shipping (Accident Reporting and Investigation) Regulations 2012 shall be the prevention of future accidents through the ascertainment of its causes and circumstances. It shall not be the purpose of an investigation to determine liability nor, except so far as is necessary to achieve its objective, to apportion blame.”

NOTE

This report is not written with litigation in mind and, pursuant to Regulation 14(14) of the Merchant Shipping (Accident Reporting and Investigation) Regulations 2012, shall be inadmissible in any judicial proceedings whose purpose, or one of whose purposes is to attribute or apportion liability or blame.

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GLOSSARY OF ABBREVIATIONS AND ACRONYMS

AIS	-	Automatic Identification System
bhp	-	Brake horsepower
BML	-	Boatmaster's Licence
CPA	-	Closest point of approach
DSMC	-	Domestic Safety Management Code
ENG 1	-	Seafarer medical certificate
LOA	-	Length overall
m	-	metre
MCA	-	Maritime and Coastguard Agency
MGN	-	Marine Guidance Note
ML5	-	Seafarer medical report form
MPU	-	Marine Policing Unit
MSN	-	Merchant Shipping Notice
NTM	-	Notice to Mariners
PLA	-	Port of London Authority
POB	-	Persons on board
PA	-	Public Address (system)
RNLI	-	Royal National Lifeboat Institution
RPM	-	Revolutions per minute
SCV Code	-	The Small Commercial Vessel and Pilot Boat Code
SMS	-	Safety Management System
STCW 95	-	The International Convention on Standards of Training, Certification and Watchkeeping for Seafarers (as revised 1995)
t	-	tonne
UTC	-	Co-ordinated Universal Time
VHF	-	Very High Frequency
VTs	-	Vessel Traffic Services

TIMES: All times used in this report are UTC+1 unless otherwise stated.

SYNOPSIS

At 1414 on 17 July 2014, the passenger vessel *Millennium Time* collided with the motor tug *Redoubt* and its towed barges between Blackfriars Road Bridge and Waterloo Bridge on the River Thames, London. Both vessels were damaged and nine passengers sustained minor injuries. There was no pollution.

The collision occurred after *Millennium Time* turned unexpectedly towards *Redoubt*, which was travelling in the opposite direction. The turn was not deliberately initiated by the movement of the helm and most likely resulted from the unintended application of port rudder, and/or the influence of the effects of hydrodynamic interaction. *Millennium Time*'s mate was on the helm and he initially assessed that the tug was turning towards the passenger vessel. Consequently, he did not take avoiding action for several seconds. By then, the vessels were too close for the collision to be prevented.

Contributing factors on board *Millennium Time* included:

- The wheelhouse was not fitted with a rudder angle indicator and the vessel was difficult to keep on a steady heading.
- The mate did not hold a boatmaster's licence and his employment as the helmsman was contrary to the operator's instructions.
- The master was providing a sightseeing commentary to the passengers and did not effectively supervise the mate on the helm.
- Problems associated with the steering system reduced its effectiveness in meeting the navigational challenges on the River Thames.
- The risk of hydrodynamic interaction was increased off the Coin Street moorings where the navigable water available had been reduced by *Redoubt* being 15m south of its usual track.

Following the collision, the master quickly secured the vessel alongside a moored barge but the crew appeared to be unable to effectively manage the passengers on board and the vessel's watertight integrity was not checked. The crew were also unaware of the number of passengers on board.

In August 2014, a recommendation was made to the Maritime and Coastguard Agency intended to ensure that the procedures and control measures adopted on board *Millennium Time* and its sister vessels were sufficient to enable the vessels to navigate safely on the Thames. Since then, the vessels' owner and operator, City Cruises, has replaced the vessels' steering systems.

This report contains a recommendation to the Maritime and Coastguard Agency which seeks to expedite legislation which will restrict the use of grandfather rights for existing domestic passenger vessels. The Port of London Authority has been recommended to assess the impact on safe navigation of the Coin Street moorings and City Cruises has been recommended to take action designed to improve the safe navigation of its vessels.

The Maritime and Coastguard Agency and the Passenger Boat Association have been jointly recommended to explore the use of technology to improve the accuracy of the passenger counts on passenger vessels operating on the River Thames.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF *MILLENNIUM TIME*, *REDOUBT* AND ACCIDENT

SHIP PARTICULARS		
Vessel's name	<i>Millennium Time</i>	<i>Redoubt</i>
Flag	UK	UK
Classification society	N/A	N/A
IMO number	N/A	N/A
Type	Passenger vessel	Tug
Registered owner	City Cruises Plc	Cory Environmental Ltd
Manager(s)	City Cruises Plc	Cory Environmental Ltd
Construction	Steel	Steel
Length overall	31.58m	22.65m
Registered length	N/A	N/A
Gross tonnage	270	86.69
Minimum safe manning	5	N/A
Authorised cargo	N/A	N/A
VOYAGE PARTICULARS		
Port of departure	London (Waterloo Pier)	London (Belvedere)
Port of arrival	London (Tower Pier)	London (Vauxhall)
Type of voyage	Passenger service (internal waters)	Towage (internal waters)
Cargo information	N/A	Towing 3 barges
Manning	5	6
MARINE CASUALTY INFORMATION		
Date and time	17 July 2014 at 1414 (UTC+1)	
Type of marine casualty or incident	Serious Marine Casualty	
Location of incident	Kings Reach, River Thames, London	
Place on board	Bow	Port side and tow
Injuries/fatalities	None	None
Damage/environmental impact	Structural damage to port bow. Broken windows at the forward end of the passenger saloon	Indented bulwark and deck penetrations. Two towed barges were also damaged.
Ship operation	On passage	On passage
Voyage segment	Transit	Transit
External & internal environment	Fine with good visibility	
Persons on board	362 passengers, 5 crew	6 crew, 1 passenger

Image courtesy of www.simplonpc.co.uk



Millennium Time

Image courtesy of Alan Chapman



Redoubt

1.2 NARRATIVE

At 1407 on 17 July 2014, the passenger vessel *Millennium Time* departed from Waterloo Pier on the south bank of the River Thames (**Figure 1**) with 362 passengers on board. The passengers were a mix of nationalities and included groups of schoolchildren. The vessel's destination was Tower Pier and the vessel's mate was at the helm.

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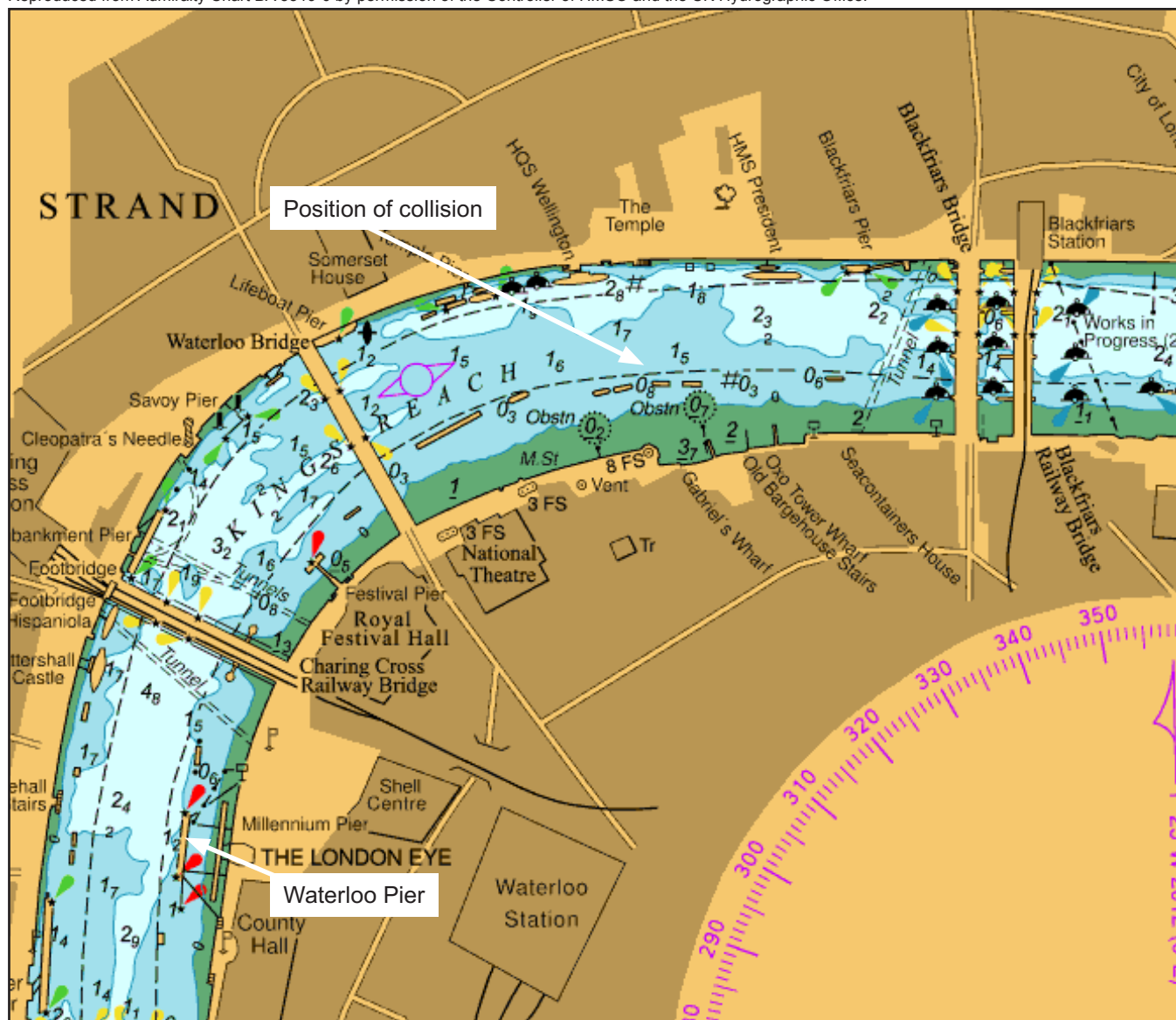


Figure 1: The River Thames from Waterloo Pier to Blackfriars Road Bridge

Millennium Time followed *Mercia*, another passenger vessel, downriver and under Waterloo Bridge. *Monsoon Clipper*, a high speed catamaran on a commuter service, was close off its port quarter (**Figure 2**). *Millennium Time* was making good a speed of 6.6 knots, about 1 knot slower than *Mercia*¹.

The passenger vessels passed under Waterloo Bridge at 1411. *Mercia*'s heading was then adjusted to starboard to follow the axis of the river to the east towards Blackfriars Road Bridge. *Millennium Time* followed approximately 40m astern.

¹ All speeds in this report are speed over the ground.

Meanwhile, *Monsoon Clipper* increased speed to over 12 knots and headed towards Blackfriars Pier on the north bank (**Figure 3**), its next scheduled stop. By now, *Millennium Time*'s skipper was providing a light-hearted sightseeing commentary using the vessel's public address (PA) system. He was standing in the wheelhouse doorway facing aft, towards the passengers (**Figure 4**).

As soon as *Millennium Time* was heading towards Blackfriars Road Bridge, the mate saw that the isophase light on the bridge was lit². The mate was aware that the light probably indicated that a tug and tow were approaching the bridge from the other side. Both *Mercia* and *Millennium Time* were now on a heading of approximately 085°, intending to pass under number 4 arch of Blackfriars Road Bridge³ (**Figure 3**) while keeping a safe distance off the barges on the Coin Street Moorings towards the south bank (**Figure 5**).

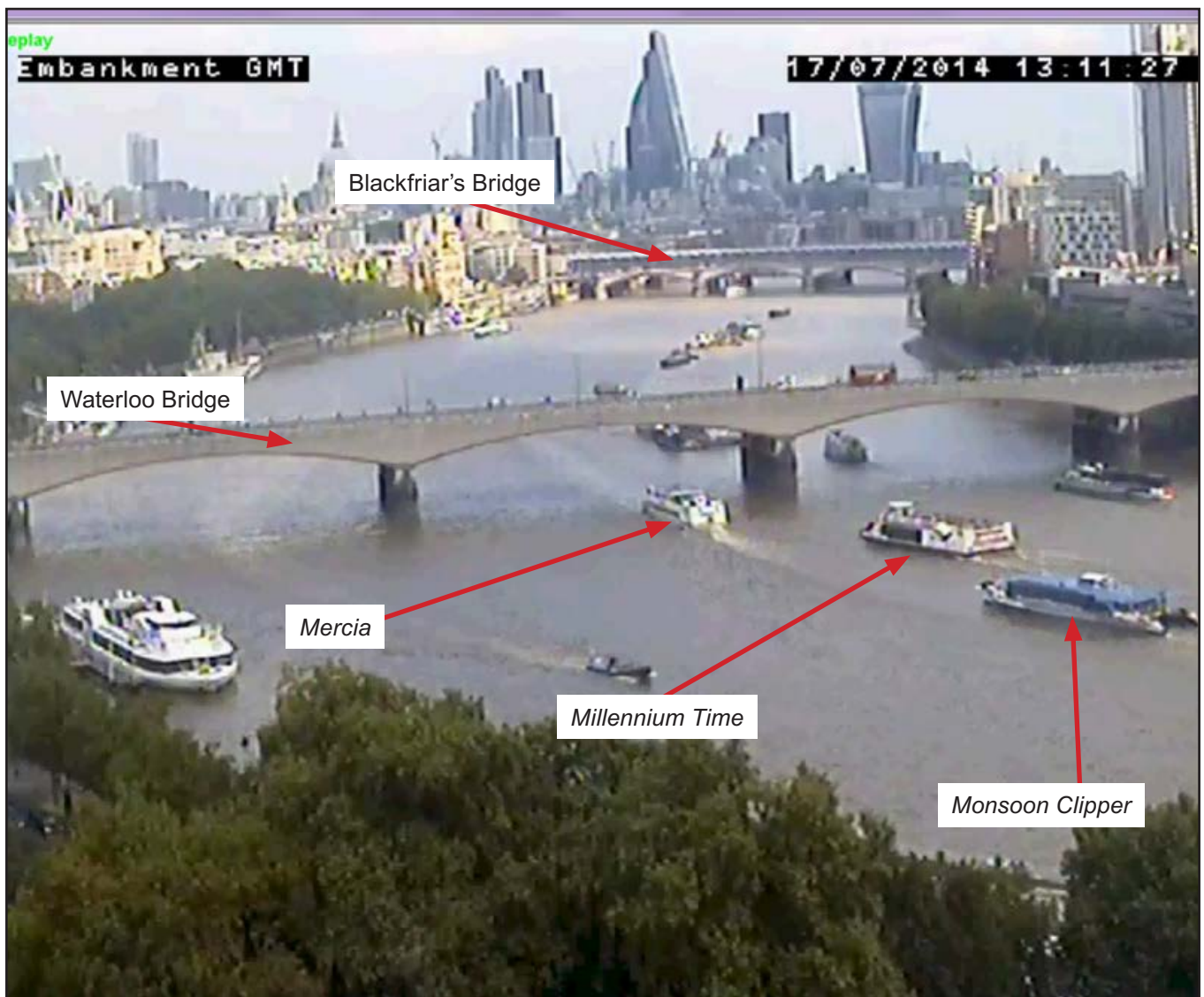


Figure 2: *Millennium Time*, *Mercia*, and *Monsoon Clipper* approaching Waterloo Bridge

² Between Tower and Wandsworth bridges, the bridges on the Thames are fitted with high-intensity white signal lights on both sides of the main navigational arches. The lights are triggered automatically at a distance of approximately 750m by a device fitted to reporting vessels to warn other river users of their approach. An isophase light (2 seconds on, 2 seconds off), indicates that a single reporting vessel is about to navigate under the bridge. A quick flashing light indicates that two or more reporting vessels are approaching.

³ Bridge arches on the Thames are numbered consecutively from the north bank.

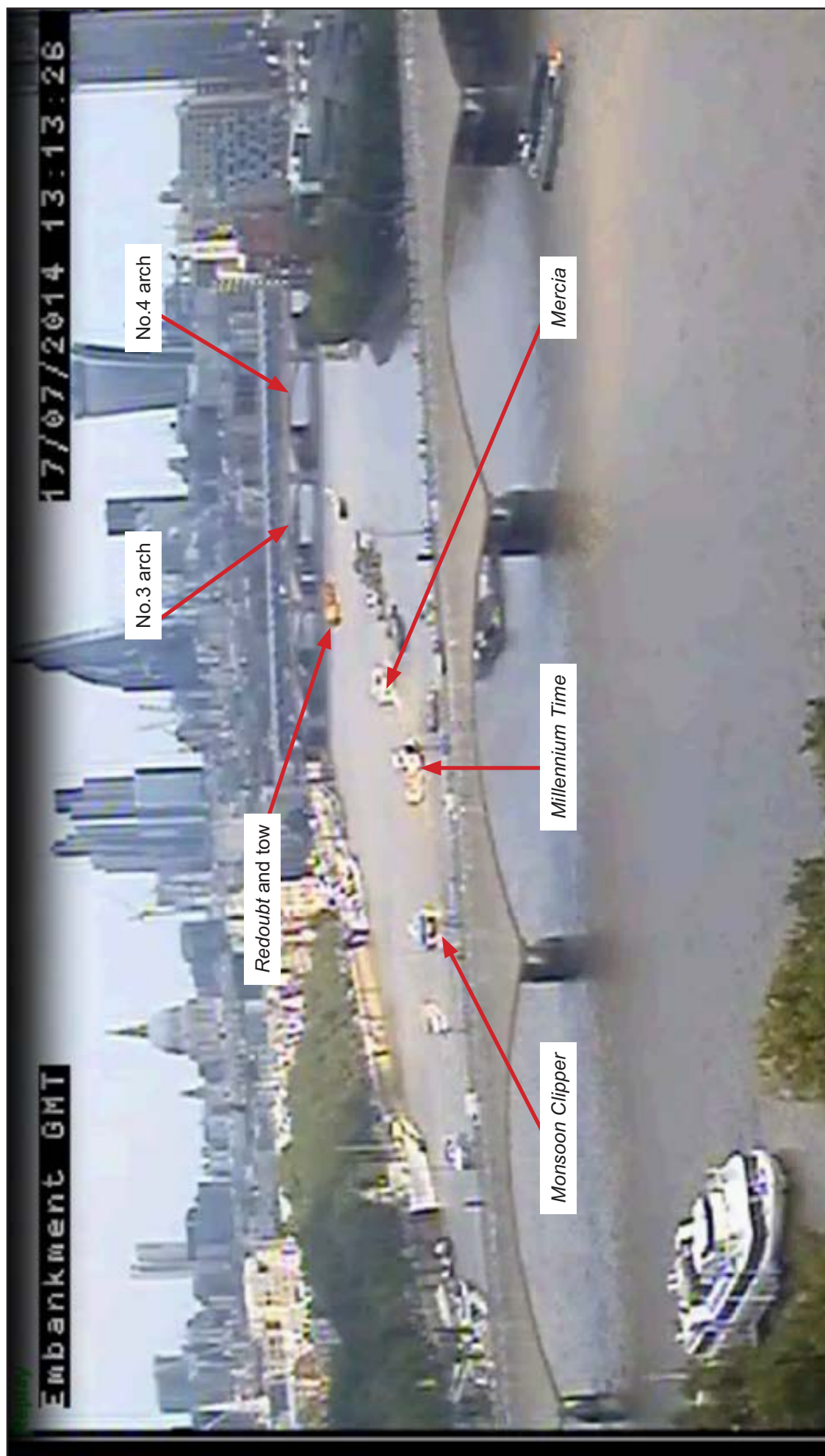


Figure 3: The vessels between Waterloo Bridge and Blackfriars Road Bridge

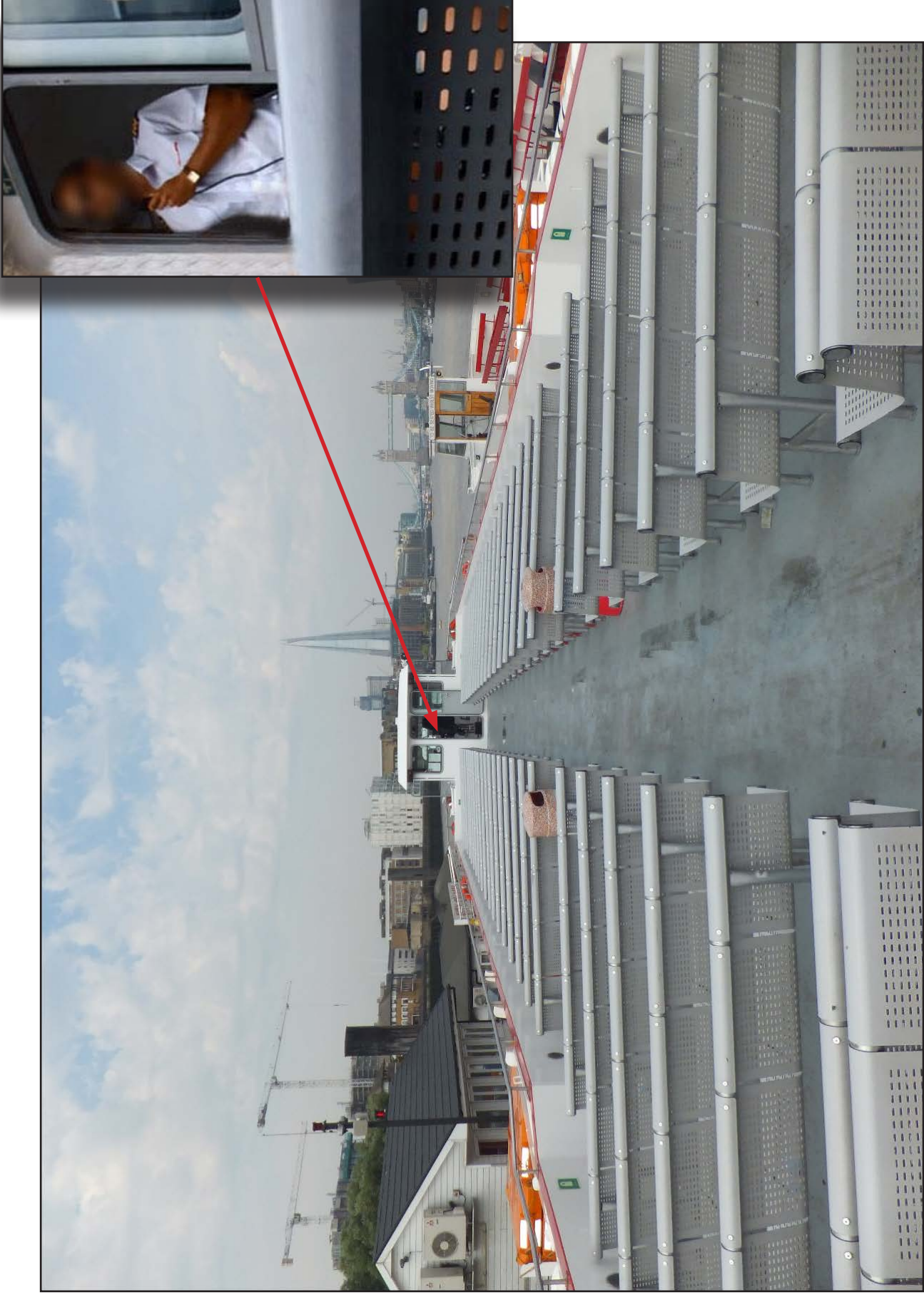


Figure 4: Top deck of *Millennium Time* showing the commentary position

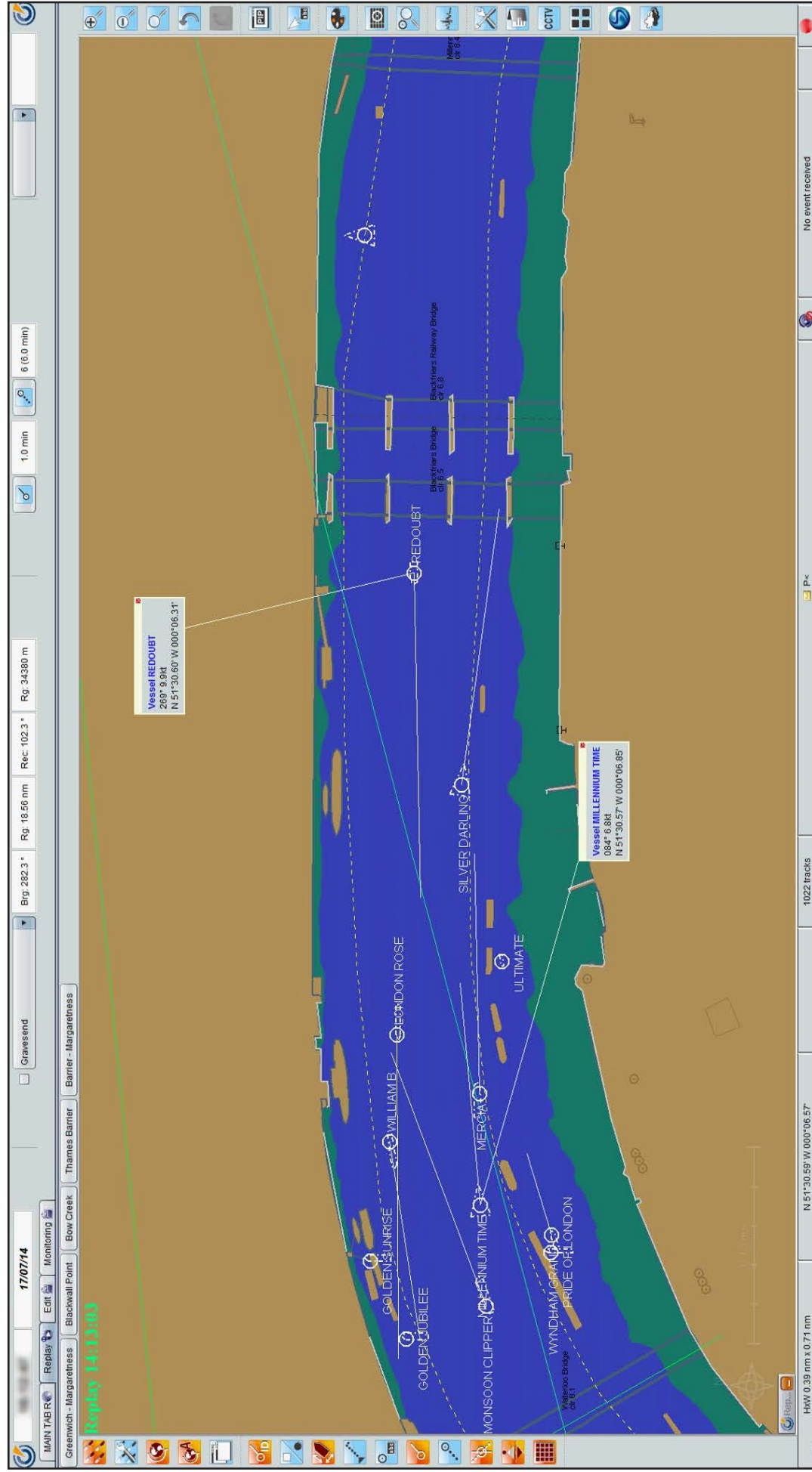


Figure 5: AIS data showing *Millennium* Time heading for Blackfriars Bridge

At 1413, the motor tug *Redoubt* with three Cringle⁴ barges on a very short tow passed under number 3 arch of Blackfriars Road Bridge (**Figure 3**). The tug and tow were on passage upriver at a speed of 10 knots. The skipper was at the helm and he was accompanied in the wheelhouse by the vessel's five crew.

Immediately after clearing Blackfriars Road Bridge, *Redoubt*'s skipper adjusted the tug's heading to approximately 265° to ensure that the towed barges astern would clear the arch abutments on Waterloo Bridge, the next bridge ahead. He intended to pass to the north of *Mercia* and *Millennium Time*, which he could see approaching off the tug's port bow.

Redoubt and its tow were seen by *Millennium Time*'s mate and its skipper. The skipper checked with the mate that he had seen the tug and then continued with the sightseeing commentary.

At 1414:02, when the distance between *Millennium Time* and *Redoubt* was approximately 110m, *Millennium Time* suddenly turned quickly to port towards *Redoubt*. Over the next few seconds, *Millennium Time*'s mate assumed that the tug had turned towards the passenger vessel and did not take any action. He then turned the helm hard over to starboard and moved the starboard engine control lever to full 'astern'. At about the same time, *Millennium Time*'s skipper saw out of the corner of his eye that the passenger vessel was about to collide with the tug and shouted to the passengers "stand fast". He then threw himself onto the deck.

By this time, *Redoubt*'s skipper had moved the tug's helm hard over to starboard to try and avoid *Millennium Time* but, at 1414.16, the two vessels collided (**Figure 6**). On impact, the tug's skipper put the helm hard to port to try and swing the towed barges clear. However, the passenger vessel struck the port swim⁵ of the southern barge in the head rank. The bow of *Millennium Time* was forced downwards and the barge rode up and hit the passenger vessel's forward saloon windows. The impact parted several lines securing the barges; it also stopped *Millennium Time* in the water. The passenger vessel was then briefly pushed astern at a speed of almost 5 knots.

1.3 RESPONSE AND RESCUE

Immediately following the collision, *Redoubt*'s skipper broadcast a "Pan Pan" urgency message on Very High Frequency (VHF radio). Extracts of the initial radio exchanges between the skipper and London Vessel Traffic Services (VTS)⁶ are at **Table 1**.

⁴ Cringle barges are named after Cringle dock near Battersea in London. The barges are flat bottomed and are designed for carrying containers. The three barges towed by *Redoubt* were arranged in two ranks (two barges in the head rank and one in the rear).

⁵ The 'swim' is the raked square end (bow or stern) of a barge that overhangs the water.

⁶ London VTS manages and oversees the safety of navigation in the Thames between Teddington to the mouth of the river's estuary. It operates from control centres in Gravesend and at the Thames Barrier Navigation Centre in Woolwich.

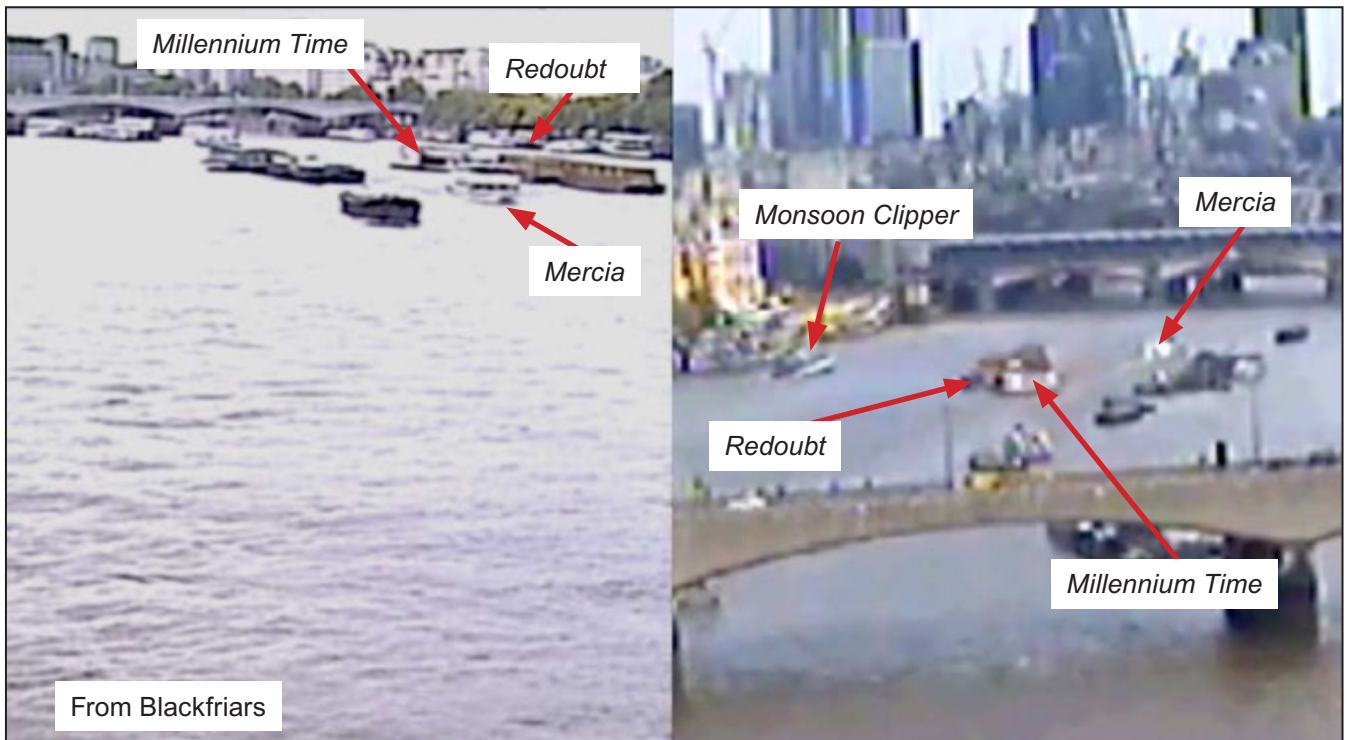


Figure 6: The collision

Time	Transmission by	Transcript
1415:01	Redoubt	<i>London VTS, this is the Redoubt. The Millennium Time has just hit our craft head-on. We are between Blackfriars and Waterloo Bridges.</i>
1415:30	London VTS	<i>Tug Redoubt, London VTS, Roger, can you give us an update; persons in the water, is the vessel sinking?</i>
1415:40	Redoubt	<i>Millennium Time is badly damaged, there has got to be some injury on board. There is no injury on board the tug. I would speak to the Millennium Time or get someone to the Millennium Time; there is probably a lot of injury on board there.</i>

Table 1: Transcript of communications between *Redoubt* and London VTS

At 1417, London Coastguard tasked a Royal National Lifeboat Institution (RNLI) lifeboat from the nearby Tower RNLI lifeboat station to attend the scene. The lifeboat was on site at about 1422 and several of its crew boarded *Millennium Time*. About 3 minutes later, patrol boats with officers on board from the Marine Policing Unit (MPU) arrived to assist. A fireboat from the London Fire Brigade and a Port of London Authority (PLA) harbour service launch with a harbourmaster on board also attended.

The RNLI and MPU noted that the passengers on board *Millennium Time* appeared to be confused and had not been prepared for evacuation. One of the customer service agents appeared to be more focused on clearing the bar area rather than dealing with the needs of the passengers. Also, the crew had not physically checked the extent of the damage to their vessel.

The police took charge and the RNLI crew quickly ascertained there were no major injuries; nine passengers had sustained cuts and bruises and one passenger was suffering with abdominal pains. MPU officers also boarded *Redoubt*.

Millennium Time's crew were unsure of the number of passengers on board. Automatic Identification System (AIS) data transmitted from the vessel⁷ indicated that 426 passengers were on board; the police counted 362. The passengers were transferred to the passenger vessel *Mayflower Garden* and were disembarked at Festival Pier where police and the London Ambulance Service were waiting.

The crews of *Millennium Time* and *Redoubt* were breathalysed for alcohol by the MPU and their mobile telephones were checked to see if they were in use at the time of the collision. The breathalyser tests were negative and no mobile telephones had been in use by the crew immediately prior to the accident.

1.4 ENVIRONMENT

The weather was fine and clear with good visibility. The wind was a gentle breeze (Beaufort force 3) from the east. Low water at Tower Bridge was 1244, 4 days after the spring tide and the height of tide was 4.2m. The predicted tidal stream was flooding at a rate of 2 knots.

A recent hydrographic survey of the river bed in the King's Reach indicated that there had been no significant changes to charted depths since previous surveys. The findings of the survey supported the published data that showed the river-bed to have a relatively flat profile (**Figure 7**). There were no outfalls or drain culverts in the area.

1.5 DAMAGE

Millennium Time sustained structural damage to its bow, and windows at the forward end of the passenger saloon were broken (**Figure 8**). The vessel was taken out of the water and took 32 days to repair.

The bulwarks on *Redoubt*'s port shoulder were pushed inboard (**Figure 9**), which resulted in four penetrations through the main deck. The repair included the removal and replacement of 5m of bulwark plating and capping. *Redoubt* was out of service for 10 days.

Damage to two of the towed cringle barges necessitated the replacement of sections of hull plating.

⁷ *Millennium Time* was one of a number of vessels that are required to carry and operate Thames AIS or AIS A when operating above Victoria Deep Water Terminal.

Figure 7: Coin St Moorings; PLA hydrographic survey

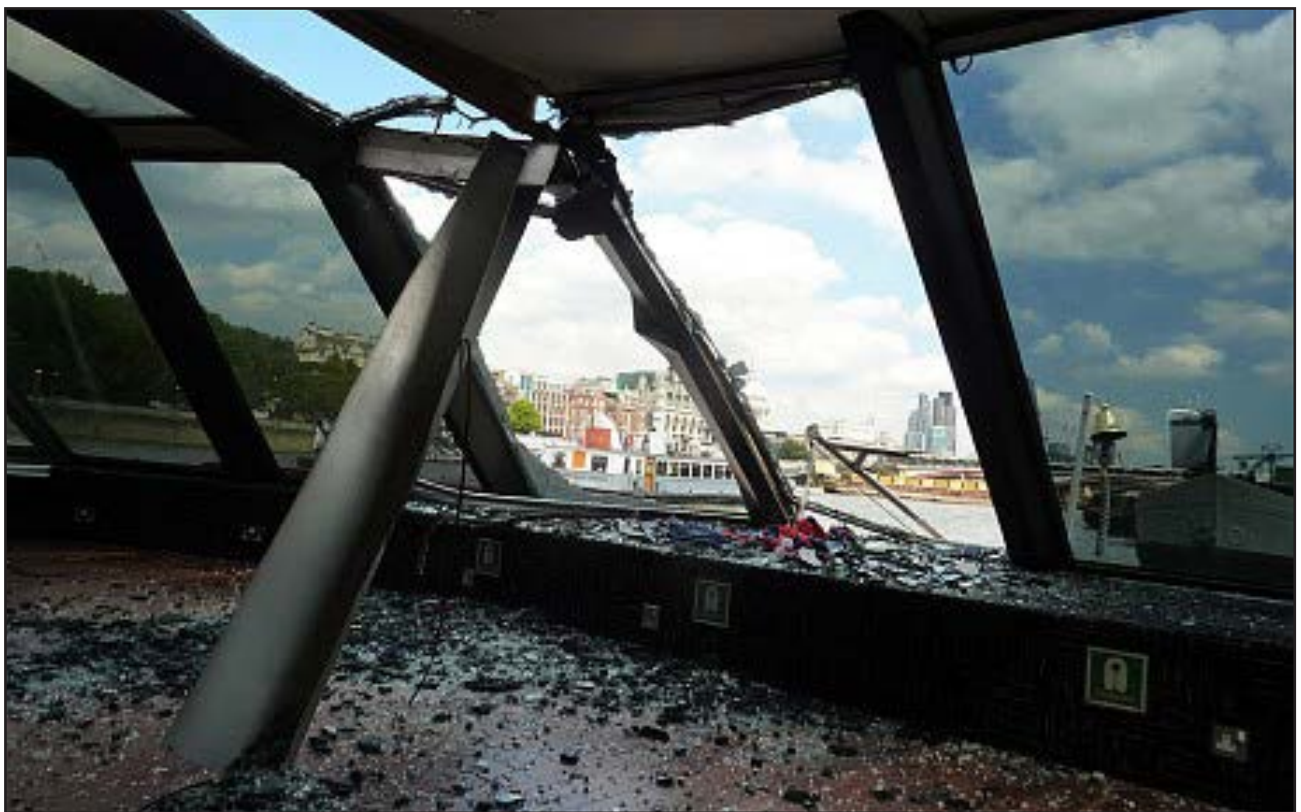


Figure 8: Damage to *Millennium Time*



Figure 9: Damage to *Redoubt*

1.6 **MILLENNIUM TIME**

1.6.1 **Design and construction**

Millennium Time was one of four M2 class steel-hulled catamarans (**Figures 10 and 11**) built by Vosper Thornycroft UK at Woolston, Southampton and owned and operated by City Cruises Plc. The other vessels were *Millennium Dawn*, *Millennium of Peace* and *Millennium City*. The catamarans, which entered service in 1999, were 31.58m in length overall (LOA) with a beam of 10.9m and a draught of 1.48m. They were among the largest passenger vessels operating on the Thames. Powered by twin 350hp Cummins main engines, the M2 class were capable of a maximum speed of approximately 9 knots. The lower deck passenger saloon was enclosed with large windows. The top deck was open. The vessels were constructed with watertight sub-divisions and fitted with 12 bilge pumps and float alarms.

1.6.2 **Certification and operation**

Millennium Time and the other M2 class vessels were certified as class V passenger vessels⁸ by the Maritime and Coastguard Agency (MCA). The vessels were authorised to carry up to 495 passengers and operated scheduled sightseeing tours between Westminster and Greenwich. The hours of operation varied with the seasonal demand and the hours of daylight available. As a result, the tours finished later during the summer months. The City Cruises' timetable for 2014 is at **Annex A**. The vessels were also used for night time river cruises and private charters for weddings, parties and corporate events.

1.6.3 **Wheelhouse and steering**

Millennium Time's wheelhouse was at the forward end of the top deck (**Figures 10 and 11**) and its equipment included:

- AIS display
- Helm/wheel
- Engine control levers
- VHF radio
- Magnetic compass (which was not functioning)
- River guide control panel
- Public address system
- Fire and bilge alarm panel.

⁸ A Class V passenger vessel is permitted to operate on inland waters Category A, B or C. The River Thames at King's Reach was Category C: *Tidal rivers and estuaries and large, deep lakes and lochs where the significant wave height could not be expected to exceed 1.2 metres at any time.*

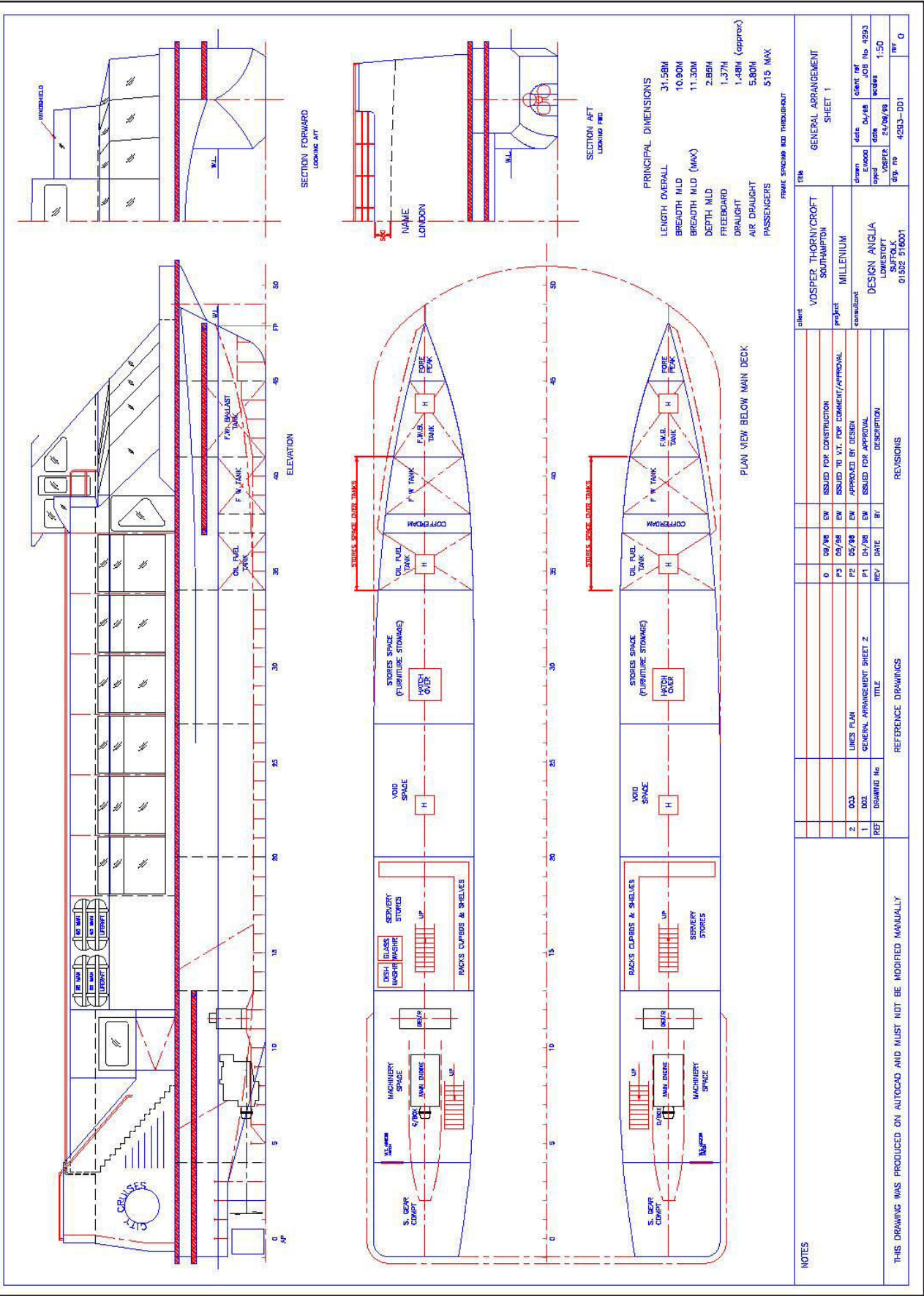


Figure 10: Millennium Time; general arrangement



Figure 11: Millennium Time; layout

Millennium Time's steering system was manufactured by Whitlock, which predominantly provided steering systems for yachts. The manufacturer was acquired by Lewmar in March 2000. *Millennium Time's* steering system (**Figure 12**) comprised:

- a large hand-operated wheel (**Figure 13**) which turned;
- a bevel gearbox⁹ (Box 1), connected to;
- a rod through a deck bearing and two universal joints, to;
- a bevel gearbox (Box 2), connected to;
- a 24m rod with seven universal joints into;
- a bevel gearbox (Box 3), connected to;
- a rod with two universal joints, into
- a bevel gearbox (Box 4), which split the drive. The system continued on each side of the vessel to;
- a rod with two universal joints into;
- a bevel gearbox (Box 5/Box 6), connected to;
- a rod with four universal joints and a through-deck bearing to;
- a reduction gearbox, controlling;
- a steering arm connected to a rudder.

Each rudder was 102cm high x 71.5cm wide, manufactured from 14mm steel plate (**Figure 14**). The rudders were not counterbalanced.

The steering system was obsolete and many spare parts had to be manufactured to order. Contractors acting on behalf of City Cruises had discussed replacing the rod and gear system with a hydraulic system with the MCA in 2009. A replacement hydraulic steering system was approved by the MCA in January 2014 and funding was authorised by City Cruises' management board 1 month later. At the time of the accident, none of the M2 class vessels had been fitted with a hydraulic steering system.

1.6.4 Maintenance

The M2 class catamarans were dry docked biennially. During the dry dock period, the vessels underwent condition-based maintenance during which critical equipment was refurbished or replaced. Any deficiencies identified at survey by the MCA were addressed during the dockings.

Millennium Time was last dry docked in January 2014. During the docking, the refurbishment of the steering system included the replacement of several bevel gears, bearings and seals. Neither the steering system nor the vessel's propulsion system had any known defects at the time of the accident.

⁹ Bevel gears are most often mounted on shafts that are 90° apart, but can be designed to work at other angles.

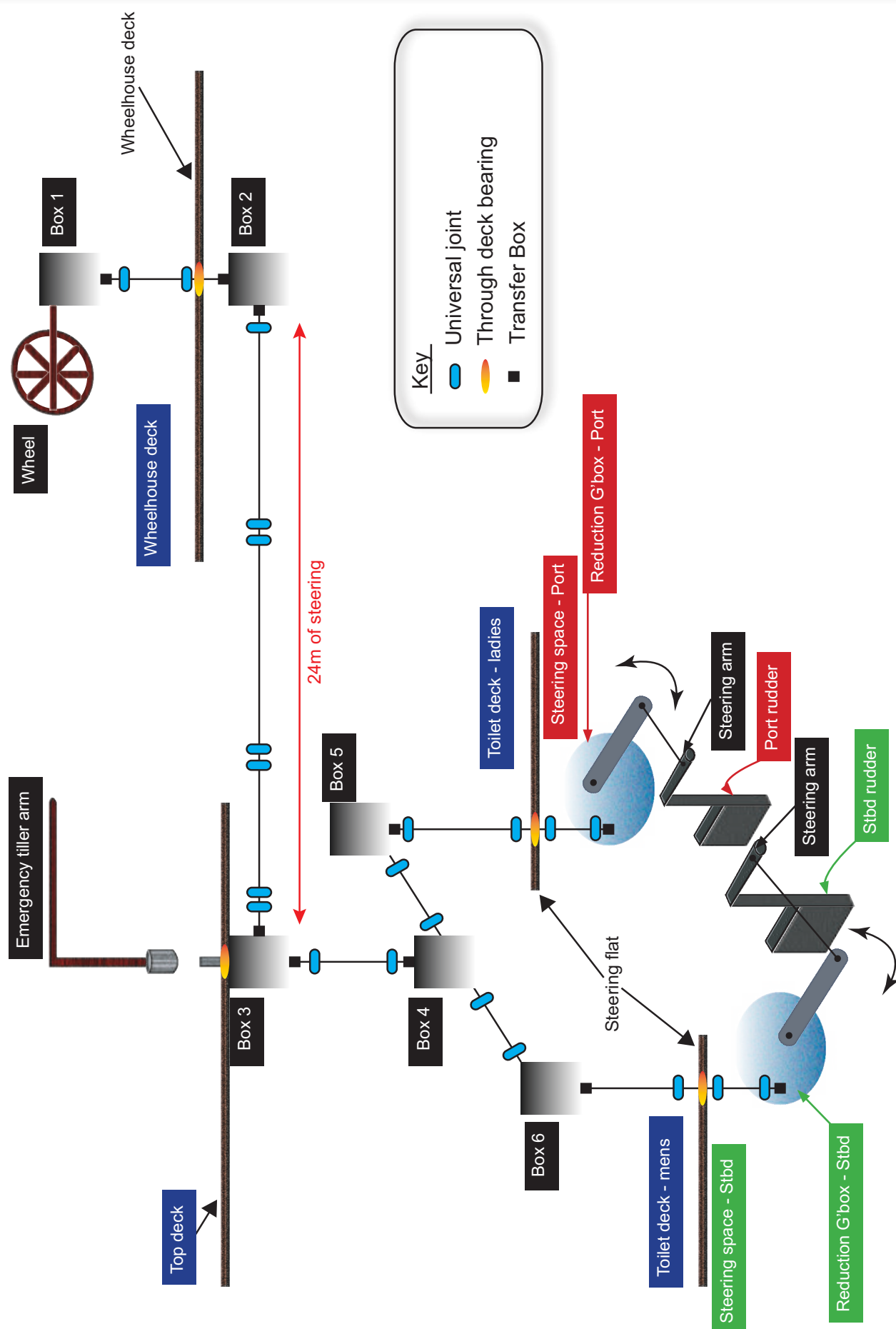


Figure 12: Millennium Time; steering system schematic



Figure 13: *Millennium Time*; steering wheel

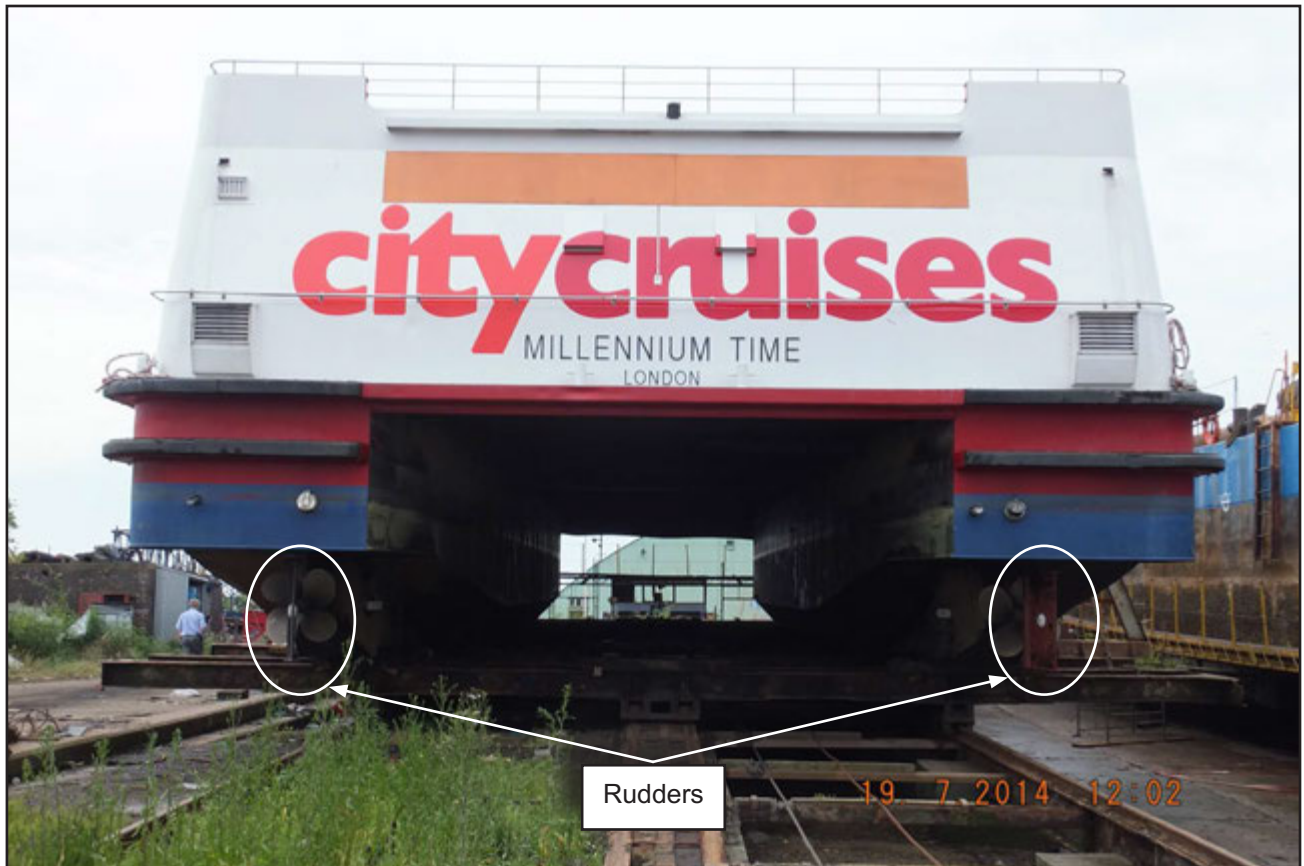


Figure 14: *Millennium Time*; rudders

1.6.5 Crew

When operating on the Thames, *Millennium Time* was approved by the MCA to carry 495 passengers by day and 493 at night, with a minimum number of five and seven crew respectively. These numbers were reflected on the Passenger Vessel Safety Certificate.

Millennium Time's crew members were recruited directly by City Cruises and comprised a master and a mate and three customer service agents. The master and mate were engaged on permanent contracts; the customer service agents were employed on seasonal contracts. The crew worked a routine of 3 days on followed by 2 days off. Each shift started at approximately 0800 and usually ended at 1700 unless the vessel was chartered for a night cruise when the crew would work into the evening.

The master was 58 years old and was a UK national. He had worked on the Thames since 1977, starting as an apprentice and then gaining the freedom of the river as a waterman. The master had worked for City Cruises for 8 years, the last 7 years of which he had served as mate then master on board the M2 class vessels. He held a Boatmaster's Licence (BML), Tier 1 Level 2 with local knowledge and large passenger vessel endorsements¹⁰. The master had completed STCW 95¹¹ training in sea survival, fire-fighting and first-aid. He had also completed training in crowd management.

The mate was 69 years old and was also a UK national. He had also spent much of his working life on the Thames as a waterman and had worked on board the M2 class since they started service in 1999; he joined *Millennium Time* in February 2014. The mate had completed a BML theory course in 2009 and he also sat the local knowledge examination in 2010. However, the mate had not completed the BML practical examination. Therefore, in common with many other mates employed by City Cruises, he did not hold a BML. There is no record of *Millennium Time*'s mate having completed a practical boat-handling assessment during his employment with City Cruises. The mate had completed STCW 95 training in sea survival and fire-fighting.

The customer service agents were Nigerian, Romanian and British nationals. The agents were issued with a 'competent crew training record' and had completed a 1-day induction with City Cruises that included health and safety training. Two of the agents had also completed a 1-day first-aid course delivered by the Royal Yachting Association and two had taken part in a collision drill in August 2014. No records were available of the training received by customer service agents to comply with the additional 'competent crew' training requirements for personnel serving on domestic passenger vessels (see paragraph 1.13.2).

1.6.6 Sightseeing commentary

Millennium Time was fitted with a 'Tourmaster System', a computerised, multilingual, river guide accessible through headphones. However, in common with other M2 vessels, it was usual for either the mate or the master to give a light-hearted verbal commentary that included their local knowledge of the Thames and its history. The

¹⁰ A Boatmaster's Licence is required for the masters of passenger ships (carrying more than 12 passengers) and non-passenger vessels of 24m and over operating on inland waterways or in limited coastal areas. See paragraph 1.14.

¹¹ The International Convention on Standards of Training, Certification and Watchkeeping for Seafarers, 1978, as amended, details qualification standards for seafarers on seagoing merchant ships.

passengers were invited to put a 'tip' in a bucket on leaving the vessel to show their appreciation of the commentary. The money taken was shared between the master and the mate.

1.6.7 Passenger count

The passengers were counted on and off *Millennium Time* using a hand-operated tally counter. When the vessel arrived at Westminster Pier before the collision, transmissions from its AIS indicated that there were 426 persons on board (POB). The number remained at 426 when the vessel departed from Westminster Pier and again from Waterloo Pier. Passengers embarked and disembarked at both Westminster and Waterloo Piers. Anecdotal evidence indicates that the use of hand-operated tally counters for counting large numbers of passengers on and off Thames passenger vessels is not always accurate.

1.7 CITY CRUISES

1.7.1 Overview

City Cruises was based at Cherry Garden Pier in south-east London and operated 19 passenger vessels that carried over 2 million passengers during 35,000 sailings annually. The company employed over 180 staff and also operated a small fleet of support vessels and barges.

Since 2008, City Cruises' operation of its vessels had met the requirements of the Domestic Safety Management Code (DSMC)¹².

1.7.2 Wheelhouse manning

In spring 2012, three City Cruises' vessels made contact with bridges within a 2 week period. These included:

- On 23 March 2012, *Millennium City* made contact with an abutment on Southwark Bridge. The mate was at the helm while the skipper was making a public broadcast on the public address (PA) system at the back of the wheelhouse.
- On 6 April 2012, *Millennium Time* was turning in the river in order to berth into the tidal stream. During the manoeuvre, the person at the helm misjudged the turn and the vessel was set onto Westminster Bridge.

Following its investigation of the accident involving *Millennium City* the PLA issued recommendations to City Cruises and the MCA, including:

(to City Cruises)

- *Cease the practice of the Master undertaking the commentary unless a Mate is provided on board who is fully qualified to have conduct of the vessel as an unsupervised watchkeeper.*
- *Consider options for improving the Master's situational awareness in the event of the Master undertaking the commentary to ensure Masters can fulfil their responsibilities; this may include ceasing the commentary around high risk areas and when manoeuvring for piers or through bridges.*

¹² See paragraph 1.14.3

- *Consider implementing a more robust and formal training and evaluation system for Mates to ensure they are competent to undertake the duties and responsibilities required of persons working in the capacity of a Mate aboard City Cruises vessels.*
- *Emphasise the need for the Master and Mate of the vessel to be fully aware of the number of POB at all times. [sic]*

(to the MCA)

- *Consider reviewing the qualification requirements for the Mate of a Class V Passenger Vessel to hold a full Tier 1 Level 2 Boatmasters' Licence with appropriate endorsements.*

On 7 April 2012, City Cruises issued two operational memoranda:

Operational memorandum 11/2012 - *"Qualified personnel at the helm"*, which stated:

"With immediate effect and until further notice, any City Cruises vessel used for the purpose of carrying passengers and in normal operation, must have a BML licence holder at the helm at all times when between Cherry Garden Pier and Westminster Pier. There are no exceptions to this rule while this memorandum is in force."

Operational memorandum 12/2012 - *"Delivery of commentary by the Master"*, which stated:

"It is the primary and principal responsibility of the Master of a vessel to ensure the safe navigation of his vessel at all times. He may delegate the driving of that boat to his mate at certain times and as long as it does not conflict with Operational Memorandum 11/2012. He must however be fully aware of his vessel's position, direction and passage plan at all times regardless of the fact that he is delivering a verbal commentary. This is particularly crucial when passing under bridges or calling at the piers."

City Cruises required its mates who did not hold a BML to complete an internal boat-handling assessment before they were allowed to steer a vessel through the bridges. The assessment required the mates to demonstrate local knowledge in addition to boat-handling skills. City Cruises did not require mates who did not hold a BML to meet a prescribed standard of medical fitness.

1.7.3 Terms of reference

City Cruises' safety management system (SMS) contained terms of reference for key personnel, including masters, mates and customer service agents. The customer service agents were responsible to the master for all matters concerning operations and safety. In particular, agents were required to:

- *Perform, in the event of an emergency such duties as required by the Captain. [sic]*

In the customer service agents job description, their main purpose was described as:

- *To ensure high standards are achieved in the safety and sales on board the vessel as well as the customer service experience of the passengers.*
- *Provides our visitors with the best service possible in terms of bar sales, customer relations and providing information about City Cruises services. [sic]*

1.7.4 Training and emergency preparedness

City Cruises' SMS required new joiners and personnel undertaking a new duty on a vessel to complete a familiarisation programme. It also required masters to be issued with a daily brief before commencing their duty that detailed the qualifications and training status of the crew allocated to their vessel.

City Cruises provided each of its vessels with an emergency procedures guide. The guide's checklists covered abandon ship, bomb threat, collision, grounding, fire, flood, man overboard, loss of power or steering, medical emergency, restricted visibility, violent or abusive behaviour and suspicious object or bag. The checklist for 'collision' is at **Annex B**.

City Cruises also allocated days on which emergencies were to be drilled. The SMS required all drills to be recorded in its vessels' daily logbook and in a master training logbook held ashore.

1.8 **REDOUBT**

1.8.1 Design and construction

Redoubt was one of four tugs designed and built by Damen Shipyards for Cory Environmental Ltd, specifically for operation on the Thames taking into account the river's bridges and shallow waters. The other tugs were *Recovery*, *Resource* and *Reclaim* and all entered service in 2010. The tugs were 22.65m LOA and were powered by twin Caterpillar engines producing 1,200 brake horsepower (bhp) and a bollard pull of 17 tons. *Redoubt*'s wheelhouse was equipped with a horseshoe-shaped console and a small recreation area aft. The visibility from the wheelhouse was good all round.

1.8.2 Certification and operation

Redoubt and its sister vessels were certified to operate under the *Small Commercial Vessel and Pilot Boat (SCV) Code*¹³ with a permitted area of operation up to Category 3 (20 miles from a safe haven). The tugs were managed and operated by Cory Riverside (Thames) Ltd and were used to tow purpose-built Cringle barges carrying compressed and containerised refuse from the City to an 'Energy from Waste' plant at Belvedere (near Crossness) in the lower reaches of the river.

¹³ The SCV Code was published in 2004 by the MCA as an annex to Marine Guidance Note (MGN) 280 (M): Small Vessels in Commercial Use for Sport or Pleasure, Workboats and Pilot Boats – Alternative Construction Standards. The SCV Code (frequently referred to as the 'harmonised Code') was intended to supersede the existing codes of practice (Blue – Small Commercial Sailing Vessels, Yellow - Small Commercial Motor Vessels, Brown – Small Workboats and Pilot Boats, and Red – Small Vessels in Commercial use for Sport or Pleasure Operating from a Nominated Point of Departure). The SCV Code has not yet been enabled but it is accepted as an equivalent alternative standard to the existing codes of practice.

Barges with loaded containers were towed from the city downriver on the ebb tide and barges with empty containers were towed upriver on the flood tide. Each Cringle barge had a capacity of 20 containers.

The dimensions of the barges were:

- Length 34.7m
- Breadth 7.5m
- Displacement (empty) 106t
- Displacement (laden) 250t

The tugs routinely towed three barges: two in the head rank and one in the second rank (**Figure 15**). When operating in the upper reaches of the river, the tow ropes between the tugs and the barges were set very short, so that tug and tow were almost a composite unit. In this configuration, the length of tug and tow was almost 100m. The towing arrangement was in accordance with the *PLA Code of Practice for Craft Towage Operations on the Thames*.

When passing between Blackfriars Road Bridge and Waterloo Bridge on a flood tide, it was usual practice for the skippers to manoeuvre to the south of the river's centreline. This was done to allow the barges, which tended to be offset to the north of the tugs, to clear the buttresses on Waterloo Bridge. Such manoeuvring should have been well known to holders of a valid local knowledge endorsement and was usually accounted for by river users.

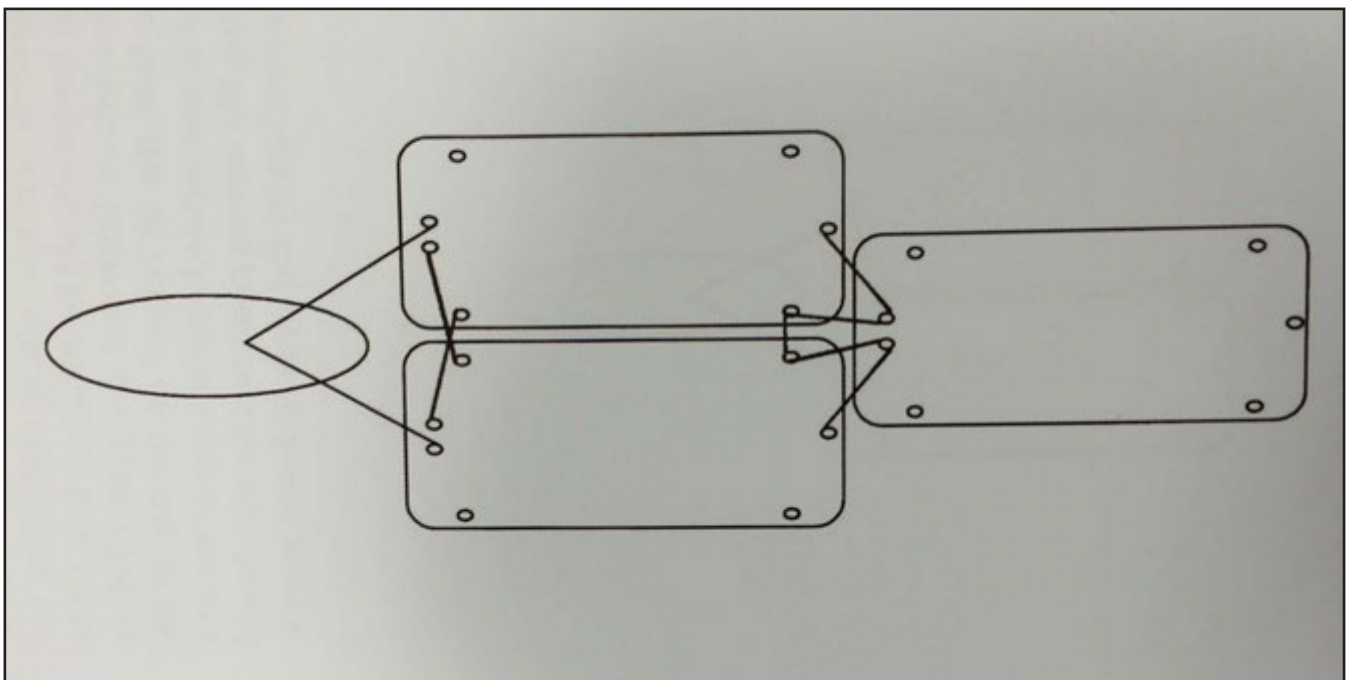


Figure 15: Towing arrangement (PLA guidance)

1.8.3 Crew

Redoubt's crew comprised a master, mate, engineer and three lightermen. The crew worked 12-hour shifts in a 3 days on followed by 2 days off rotation. The working day commenced between 0600 and 0900 depending on the work programme and the tidal conditions.

The master was a 56 year old UK national. He was a Thames waterman and had extensive experience in towage operations both on the Thames and abroad. The master had worked for Cory Environmental Ltd since 1999 as a lighterman, mate and relief master. He was promoted to master in 2012 and held a BML, Tier 1 Level 2 endorsed for cargo, towing/pushing, local knowledge and passenger operations. The mate and lightermen also held BMLs.

On the day of the accident, *Redoubt* was carrying a friend of one of the lighter-men as a passenger. He had no nautical knowledge or experience and was on the foredeck at the time of the collision.

1.9 CORY RIVERSIDE (THAMES) LTD

Cory Riverside (Thames) Ltd managed its vessels through an SMS in accordance with PLA's *Code of Practice for the Management and Operation of Commercial vessels on the Thames 2013*¹⁴. In common with several commercial vessel operators on the Thames, it was the company's policy for all its crews, apart from its engineers, to either hold a BML or be working towards one.

1.10 INSPECTION, TRIALS AND MEDICAL EXAMINATION

1.10.1 Post-accident inspection

Immediately following the collision, MCA surveyors witnessed functional tests of *Millennium Time*'s main engines and its steering system. No defects were found. A fuel sample was also analysed but no irregularities were identified.

The vessel's steering system was examined by MAIB inspectors. The large steering wheel (**Figure 13**) required almost five turns to move the rudder from hard over to starboard to hard over to port and vice versa. The wheel had no markings to indicate when the rudder was amidships and the wheelhouse was not fitted with a rudder angle indicator.

Considerable lag was identified between the initial movement of the wheel to the movement of the rudders. The wheel was turned 3 spokes (approximately 1/3 of a complete turn) before any movement of the rudders was seen.

1.10.2 Manoeuvring trials

Following repairs to *Millennium Time*, MAIB inspectors conducted steering trials on board the vessel on 19 August 2014. The trials were held near Gravesend on the lower reaches of the Thames and were intended to determine the vessel's handling characteristics and the effectiveness of its steering system. Runs were conducted

¹⁴ The purpose of the Code is to establish a common standard for the safe operation of freight and general commercial service vessels operating on the tidal Thames.

in daylight into and against the tidal stream with the engine speed adjusted to give a speed of approximately 7.5kts. The wind speed was between 15 and 17kts. The trial was attended by representatives of City Cruises, the PLA and the vessel's insurers.

During the trial the MAIB inspector identified that:

- Considerable force was required to turn the wheel hard over to both port and starboard while the vessel was making way with the engines running at their normal speed.
- It was difficult to keep the vessel on a straight course.
- The forward and elevated position of the wheelhouse meant that the helmsman had no visual reference, such as the bow tip, to accurately judge the rate of turn. The rate of turn was frequently assessed by looking aft and watching the movement of the stern.
- The helmsman did not know the position of the rudder because no rudder indicator was fitted.

A second manoeuvring trial was conducted on 5 September 2014 on the Kings Reach when the tidal conditions were similar to the conditions at the time of the collision. Prior to the trial, City Cruises fitted a rudder indicator at a cost of about £3000 together with a new magnetic compass in *Millennium Time's* wheelhouse. In addition to MAIB inspectors, representatives from City Cruises, the PLA, the MCA and the vessel's insurers also attended.

The purpose of the trial was to try and determine, on an opportunity basis, how the local conditions and variations in *Millennium Time's* propulsion might have influenced the vessel's movement. The possibilities of interaction¹⁵ and a loss of power in the port engine were also considered and simulated where possible.

Several runs were conducted from Waterloo Bridge to Blackfriars Road Bridge during which the port engine speed was reduced in increments of 25%. During the last of these runs, the port engine was reduced to almost idling speed and the rudders were amidships when *Millennium Time* turned 40° to port in 30 seconds (the starboard engine was running at its normal operating power).

Two additional trials were conducted to establish if the interaction from a passing Thames Clipper vessel¹⁶ had any discernible effect on *Millennium Time's* heading. During the first of these trials, *Millennium Time's* helm was left unattended. As the Thames Clipper vessel passed on the port side at a distance of approximately 15m and then drew clear, *Millennium Time* turned to port at a steadily increasing rate which reached 87.1° per minute before the helmsman intervened and regained control (**Figure 16**). During the second trial, in which a helmsman remained at the wheel, the same effect was seen but the helmsman was able to maintain a steady course by using between 5° and 10° of counter helm.

¹⁵ Interaction or hydrodynamic effect is a phenomenon which can be observed as bank effect, squat or just an unexpected movement between passing or overtaking vessels. There are several factors that will affect the extent of interaction, including speed of the vessel, hull form, depth of water, and proximity of obstructions and other vessels.

¹⁶ Thames Clippers operates 13 high speed catamarans running a daily scheduled passenger service between Waterloo and QE2 Pier via Canary Wharf and Greenwich. The majority of the fleet are 38m in length with a maximum speed of 30 knots and a passenger capacity of 220.

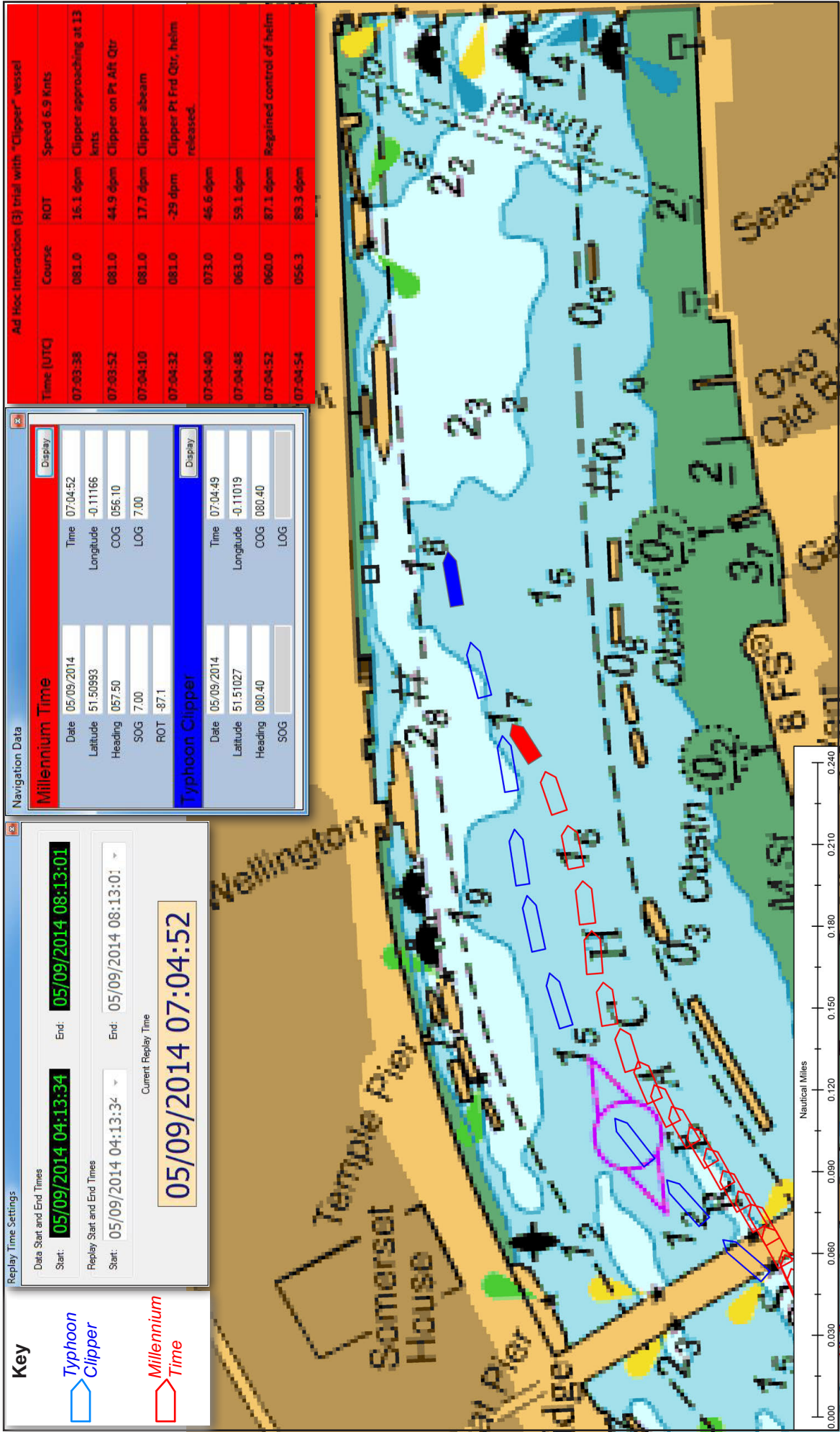


Figure 16: Effect of interaction during manoeuvring trials

Over the course of the trials, *Millennium Time* passed close by to the Coin Street moorings on 14 occasions. No effects of hydrodynamic interaction were observed during these passes.

It was observed that the fitting of a rudder indicator assisted the helmsman to steer a steady course. Therefore, the vessel appeared to be more directionally stable than seen during the previous trial in August.

The steering trials in both August and September demonstrated that the steering was heavy when both engines were running at normal operating speed. However, *Millennium Time* responded quickly when more than 10° of helm was applied.

1.10.3 Medical examination

On 3 October 2014, *Millennium Time*'s mate was examined by an MCA approved doctor for assessment against the ENG1 medical standard¹⁷. The examination did not identify any medical conditions that were relevant to the mate's actions at the time of the accident.

1.11 PLA INVESTIGATION

The PLA's investigation report of the collision between *Millennium Time* and *Redoubt* concluded the causal factors on board *Millennium Time* to be:

- Inattention by the helmsman – “*either by not paying sufficient attention or not able to react quickly enough to the expected interaction.*”
- The manoeuvrability of *Millennium Time* - due to “*the inherent issues in the steering system of the Millennium Time.*”
- Crew competency - “*the Mate was placed into the position of having unsupervised conduct of the vessels due to the Master being preoccupied undertaking the commentary.*” [sic]

The investigation also identified a number of other safety issues on board *Millennium Time*, including:

- The master was distracted by giving a sightseeing commentary.
- The crew did not prepare the passengers for evacuation and the vessel had not been checked for water ingress. The emergency services managed the incident on board until the arrival of City Cruises' senior management.
- The mate did not hold a valid medical certificate.

The investigation also identified through the examination of historical AIS data that *Redoubt*'s track after passing under Blackfriars Road Bridge was approximately 15m to the south of the track usually followed by other tugs operated by Cory Riverside (Thames) Ltd. The investigation report, dated 14 October 2014, concluded that had *Redoubt*'s line between Blackfriars and Waterloo Bridges been more typical, it

¹⁷ The ENG1 is a seafarer's medical examination that is valid for a maximum of 2 years.

would have afforded both *Redoubt* and *Millennium Time* more time in which to take evasive action, but that it is unlikely that the collision would have been avoided. The investigation report's recommendations are at **Annex C**.

1.12 THE PORT OF LONDON

1.12.1 The River Thames

The River Thames is the longest river in England and passes through London. It is tidal from its estuary to Teddington Lock and is crossed by 29 bridges along this stretch. The river from Lambeth Bridge to Wapping Ness, in which *Millennium Time* and its sister vessels operated, is recognised to be navigationally complex with strong tidal streams, 11 bridges and a multitude of piers and moorings.

Approximately 40 passenger boat operators trade on the River Thames, resulting in around 200,000 passenger vessel movements carrying 8 million passengers each year; about 750,000 people a year commute to work using the regular shuttle services. Over 70 pleasure vessel clubs and marinas are located on the River Thames, in addition to a large number of rowing clubs. The PLA approves and monitors about 300 organised events on the river each year.

1.12.2 Port of London Authority

The PLA was established in 1909 by the Port of London Act and is the statutory harbour authority for the 95 mile tidal stretch of the River Thames. The PLA's area of responsibility is divided into two districts: Upper (from Teddington Lock to Crossness) and Lower (from Crossness to the Thames Estuary). Each of the districts is administered by a harbourmaster, both of whom report to a chief harbourmaster.

Specific requirements for commercial and leisure craft on the River Thames are detailed in the Thames Byelaws 2012. However, the PLA is empowered by the Port of London Act to issue navigational instructions to masters of vessels through General Directions¹⁸. General Directions issued by the PLA included the requirements for the carriage and use of AIS, vessel reporting and the use of the Central London Special Signal ("isophase") lights. The PLA also issues Notice to Mariners (NTM) to provide information and advice on works and events as well as changes to rules and regulations that might affect river users. NTMs may be either temporary or permanent.

1.13 REGULATION AND GUIDANCE

1.13.1 Construction standards

As a Class V passenger vessel, *Millennium Time* was required to comply with the construction standards detailed in Merchant Shipping Notice (MSN) 1699 -*The Merchant Shipping (Passenger Ship Construction: Ships of Classes III to VI(A)) Regulations 1998*. MSN 1699 does not require ships fitted with non-powered steering to have the rudder angle indicated at the steering position.

¹⁸ A vessel's master may choose not to comply with a General Direction if he can prove that he has reasonable grounds to believe that compliance would imperil his vessel or that compliance was impractical.

MSN 1823 (M) Safety Code for Passenger Ships Operating Solely in UK Categorised Waters, which was published in April 2010, introduced revised regulations for new passenger ships. The code requires that any passenger vessel operating in Category C and D waters is fitted with a rudder angle indicator.

In his Interim Report of the Thames Safety Inquiry¹⁹, Lord Justice Clarke recommends that new safety regulations should, as a general rule, be applied equally to new and existing vessels, subject to a power to grant exemptions based on practicability and the provision of satisfactory alternatives.

In February 2002, in response to Lord Justice Clark's recommendation, the MCA published MGN 218(M) Thames Safety Inquiry Recommendation 27.2 standards for new and existing ships, which stated that in principle:

- (a) All future domestic safety legislation proposed by the MCA should apply equal safety standards to new and existing vessels;
- (b) The legislation should include a power of exemption;
- (c) Exemptions will be issued where compliance with a new standard would be unreasonable (whether on the grounds of practicability or for some other reason), and where an equivalent level of safety is achieved by some means other than strict compliance with the regulations' requirements.

MGN 218(M) also states:

This policy will not be applied retrospectively – ie existing legislation will not be revised in order to conform to this principle. However, where domestic regulations are being revised for other purposes, the principle of a single set of standards should be taken into account.

When MSN 1823 came into force in May 2010, it was the MCA's intention also to introduce legislation in October of the same year restricting the 'grandfather rights'²⁰ of existing domestic passenger vessels. Proposed measures included treating existing vessels as 'new' for the purpose of applying standards if the vessel has undergone major modifications or when a vessel has not been issued with a passenger vessel safety certificate for 5 years or more. Due to changes in government priorities, the intended legislation regarding 'grandfather rights' was not introduced as planned. However, the MCA has continued to develop measures intended to restrict 'grandfather rights' in safety-critical areas such as intact and damage stability, buoyancy, fire prevention, lifesaving apparatus and wheelhouse visibility. The MCA is striving to introduce legislation incorporating these measures within the next 2 years.

1.13.2 Manning and training

Lord Justice Clarke also recommended a review of manning requirements for passenger vessels operating on the Thames to consider whether they were sufficient to ensure the safety of passengers in the event of an emergency. As a

¹⁹ In the early hours of 20 August 1989, the dredger *Bowbelle* collided with the passenger vessel *Marchioness* in Kings Reach on the River Thames. Of the 131 passengers on board *Marchioness*, 51 perished. In 1999, a public inquiry, known as the Thames Safety Inquiry into the *Marchioness/Bowbelle* disaster, was chaired by Lord Justice Clarke.

²⁰ The term 'grandfather rights' refers to the principle that vessels should only be required to comply with the standards that applied when they were built or entered service.

result, *inter alia*, the MCA published MGN 203 (M) *Crew Training for Personnel Serving on Domestic Passenger Vessels* in 2003 and MGN 290 (M) *Local Passenger Vessels: Manning* in 2005.

MGN 203 (M), states:

Competent Crew training is the minimum level of training that a person shall receive before being recognised as part of the permanent crew for the purpose of the minimum manning recorded on the Passenger Certificate or Safe Manning Document.

Competent crew training broadly matches the STCW 95 training courses available for merchant seafarers but may be delivered at company level providing it is fit for purpose. The training includes the identification of safety equipment and action to be taken in the event of an emergency. Records of training are required to be documented and maintained by the owner/operator as part of the SMS.

MGN 290 (M) explains the statutory requirement for the operators of, among other vessel types, Class V passenger vessels to ensure that sufficient crew are on board for their safe and efficient operation. It also gives guidance on how operators may determine, and the MCA will assess, a safe level of manning, setting out the manning levels necessary for dealing with an emergency and the factors to be taken into account. The MGN includes:

2.2 Safe manning of the vessel is the operator's responsibility, based on their assessment of risks and the characteristics of the vessel.

2.6 The minimum crew for a passenger ship is two - ie the master plus one other.

2.7 If asked to do so, the MCA will consider a sliding scale of crew depending on the number of passengers carried. The passenger certificate may show the minimum manning levels, or allow variations for day time and night time operation or vary according to the number of passengers carried.

3.1(a) It must be possible to inform, instruct and control all passengers with the crew available.

1.13.3 Passenger count

The *Merchant Shipping (Counting and Registration of Persons on Board Passenger Ships) Regulations 1999*²¹ came into force on 1 January 2000. The regulations apply to all UK passenger vessels and to all passenger vessels in UK waters carrying 12 or more passengers. Advice in support of the regulations is provided in MSN 1794 (M+F) *Counting and registration of persons on board passenger ship*. In summary, the regulations require that an accurate record is held on board and ashore of the number of passengers on board a passenger vessel at all times. The number of passengers on board must be updated at each point of embarkation and disembarkation.

²¹ These regulations also resulted from Thames Safety Inquiry recommendations.

Thames Byelaw 2 requires the number of Persons on board (POB) to be reported via AIS. The byelaw states:

12.3 In addition, Class IV, V and Class VI passenger vessels navigating or working in the Thames between Richmond Lock and the Denton Limit must:

- a) carry and operate Thames AIS or AIS A;*
- b) carry an electronic POB reporting system which meets the minimum standards for Thames AIS, specified by the PLA in the Thames AIS Technical Requirements Specification; and*
- c) operate such POB reporting system in accordance with the procedures specified by the PLA in Procedures for the Operation of Thames AIS and Persons On Board Reporting System.*

1.13.4 Safety management

MSN 1754 (M), *The Safety Management Code for Domestic Passenger Ships of Classes III – VI (A)* provides a common standard for the safe operation of passenger ships employed in the domestic trade. It includes:

Each operator shall develop and implement safe practices which include the following:

- a health and safety protection policy;*
- procedures to ensure safe operation of ships in compliance with relevant rules;*
- lines of communication between personnel, ashore and afloat;*
- procedures for reporting accidents; and procedures for responding to emergency situations.*

1.13.5 Steering and sailing

Part 4 of the Thames Byelaws 2012 – ‘*Steering and Sailing Byelaws*’ includes:

27.1 A vessel of less than 40 metres in length navigating above Cherry Garden Pier, and a sailing vessel navigating above Cherry Garden Pier must not impede the passage of:

- (a) A vessel of 40 metres or more in length; or*
- (b) A vessel engaged in towing.*

27.1 A vessel of less than 40 metres in length navigating above Cherry Garden Pier must not impede the passage of:

- (a) A vessel of 40 meters or more in length; or*
- (b) A vessel engaged in towing.*

1.14 BOATMASTER'S LICENCE

1.14.1 General requirements

A BML is issued in accordance with the requirements set out in the Merchant Shipping (Inland Waterways and Limited Coastal Operations) (Boatmaster's Qualifications and Hours of Work) Regulations 2006. Information on the structure of the BML and the requirements set out in the regulations are detailed in MSN 1808(M) *The Merchant Shipping (Inland Waterways and Limited Coastal Operations) (Boatmasters' Qualifications and Hours of Work) Regulations 2006 – Structure and Requirements*.

The BML has two tiers: Tier 1 and Tier 2. The Tier 1 BML is a national licence which is transferable between different areas subject to local knowledge requirements. The Tier 2 BML is a local, operation-specific, qualification that restricts the holder to the waters and operation specified on the licence.

The Tier 1 BML has two levels. Level 1 is valid for Category A and B and non-linked Category C waters. Level 2 allows the holder to operate vessels anywhere within the UK's inland waterways and within limited coastal areas²².

The Tier 1 BML has a modular structure comprising:

- a main generic section (for all candidates)
- ancillary safety training (for all candidates)
- specialist endorsements (for the types of vessel or operation on which the candidate will be working)
- a local knowledge endorsement (for operations in specified areas only).

1.14.2 Generic competencies and ancillary safety training

The generic licence covers the core competencies and the boatmanship skills needed for operating in the relevant water categories. On its own, the generic licence is considered by the MCA to be a suitable qualification for operating workboats and vessels carrying up to 12 passengers.

The ancillary safety training comprises three basic safety courses: personal sea survival, first-aid and fire safety, and is an integral part of the competency requirements for the BML.

1.14.3 Specialist endorsements

A specialist endorsement is an additional requirement for BML holders that is intended for skippers on vessels engaged in certain types of operation. The specialist endorsements available are *cargo – general, oil cargoes, dredging, towing and pushing, passenger operations – general, large passenger vessel, fast craft and radar*.

²² Inland waterways means Categorised Waters A, B, C and D as defined and listed in MSN 1827; and any non-categorised inland waters. Limited coastal areas means no more than 3 miles from land and no more than 15 miles from point of departure.

1.14.4 Local knowledge endorsements

Local knowledge is defined in MSN 1808(M) as *knowledge of the features and characteristics within an area that present a hazard to safe navigation, and how to deal with them, beyond what might be expected under the BML generic skills.*

The areas for which local knowledge endorsements are required are listed in MGN 334(M) *New National Boatmaster's Licence – Local Information and Local Knowledge*, and include tidal River Thames between Putney Bridge and the eastern limit of the Thames Barrier Control Zone.

1.14.5 Medical fitness

Evidence of medical fitness is required for a first BML application. It is then required at 5-year intervals from age 45 and then annually from age 65. This evidence can be in the form of a seafarer medical report form (ML5) or a seafarer medical certificate (ENG 1) (or equivalent).

The purpose of a medical assessment is to obtain a factual report of a seafarer's medical history and present state of health, enabling a doctor to decide on the individual's fitness to navigate safely and to undertake emergency duties.

1.15 SIMILAR ACCIDENTS

Since December 2002, the four M2 class vessels operated by City Cruises have been involved in at least 17 contacts with fixed objects. One vessel also grounded.

Since 2012 there have been 12 instances of collision or contact involving class V passenger vessels in the Kings Reach area of the Thames. M2 class vessels were involved in three of these accidents.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 THE COLLISION

The collision between *Millennium Time* and *Redoubt* resulted from the passenger vessel turning to port towards the tug and its towed barges. The vessels were only 110m apart and were closing at a speed of over 16 knots on near reciprocal headings. AIS and CCTV evidence (**Figures 5 and 6**) indicates that *Millennium Time* turned 27° in 14 seconds before hitting *Redoubt*.

Prior to the turn, *Millennium Time*'s mate, who was at the helm, was aware of the proximity of the tug, which was due to pass down the passenger vessel's port side at a distance of approximately 15m. The mate did not deliberately initiate the turn by turning the steering wheel to port or by adjusting the engine control levers. Therefore, the turn or sheer must have resulted from several degrees of port rudder being set without the mate being aware, and/or the effects of hydrodynamic interaction. There was no evidence of any mechanical failure of the vessel's steering or propulsion systems.

Millennium Time's mate initially assessed that the tug was turning towards the passenger vessel and it took him several seconds to realise it was in fact *Millennium Time* that was turning. By then, although both the mate and *Redoubt*'s skipper took avoiding action, their efforts were too late to be fully effective.

2.3 RUDDER INDICATION

The absence of rudder angle indication in *Millennium Time*'s wheelhouse meant that the mate had no way of knowing the angle of the rudders. Consequently, it is possible that several degrees of port rudder had been applied when *Millennium Time* unexpectedly turned towards *Redoubt*.

The trials conducted in August 2014 highlighted the difficulty encountered in maintaining *Millennium Time* on a straight course. However, this task was made easier during the later trials in September 2014, for which a rudder angle indicator had been fitted. Although more visual references were also available in the latter trial that allowed the helmsman to steer by 'line of sight', the indicator enabled him to immediately check the angle of the rudders at all times. This removed the need for the helmsman to almost continually monitor the swing of the vessel's stern.

Millennium Time entered service in 1999 and complied with *The Merchant Shipping (Passenger Ship Construction: Ships of Classes III to VI(A)) Regulations 1998* (MSN 1699), which did not require rudder indicators to be fitted on Class V passenger vessels with non-powered steering. The revised regulations for domestic passenger vessels, introduced in MSN 1823 (M) in 2010, applied only to new vessels. Consequently, although new Class V passenger vessels operating in Category C and D waters were required to have a rudder angle indicator, existing vessels such as the M2 class were not.

Nonetheless, the benefits of the rudder angle indicator to *Millennium Time's* helmsman observed during the trials in September 2014, and the circumstances of this accident, support the case for the retro-fitting of rudder angle indication in the wheelhouses of other Class V passenger vessels. This could be done on a case by case basis, taking into account factors such as the steering arrangement, the size of the vessel and the number of passengers carried, the cost of installation and the area of operation.

2.4 INTERACTION

It is possible that *Millennium Time's* sudden sheer towards *Redoubt* was influenced to some extent by the effects of hydrodynamic interaction. The manoeuvring trials conducted in King's Reach in September 2014 showed the extent to which *Millennium Time* was potentially affected by another passenger vessel in close proximity. The passage of the vessel down *Millennium Time's* port side at a distance of about 15m initiated a change of heading of 21° in 20 seconds without any change to the rudder position (**Figure 16**). Although slower than the rate experienced immediately before the collision, this was still very significant.

Before colliding with *Redoubt*, *Millennium Time* was making good about 6.6 knots over the ground against a 2 knot tidal stream and had an under keel clearance of approximately 4.4m. As the under keel clearance was three times the vessel's draught, neither the vessel's speed nor the depth of water are likely to have adversely influenced its movement. In addition, *Millennium Time* did not experience any discernible effects of interaction when passing the Coin Street moorings during the subsequent manoeuvring trials.

Nonetheless, at the time of the sheer which led to the collision with *Redoubt*, *Millennium Time* was passing the Coin Street moorings at a distance of about 15m, *Mercia* was 90m ahead, *Redoubt* was off the port bow at a distance of 110m and *Monsoon Clipper* was overtaking a similar distance off *Millennium Time's* port beam (**Figure 6**). Therefore, this was a complex situation in terms of the hydrodynamic forces at work and it is not possible to single out a particular influencing factor. It is also very difficult to accurately quantify how the potential sources of interaction affected *Millennium Time* given the different hull forms and corresponding pressure zones of the vessels involved.

The width of the navigable water in the King's Reach is significantly reduced by the Coin Street moorings on the south bank (**Figures 2, 3, 5, 6 and 7**). Consequently, vessels frequently have to pass at close quarters and the risk of interaction is increased in this area.

In this case, AIS data indicates that *Redoubt's* track was approximately 15m further south of the tracks historically taken by similar vessels after passing Blackfriars Road Bridge. By itself, this was not unsafe, but it left only about 30m of water between the tug and the moorings for *Mercia* and *Millennium Time* to navigate. Both of the passenger vessels were required by byelaw to avoid impeding the tug and were intending to pass under number 4 arch of Blackfriars Road Bridge. Consequently, the vessels possibly passed closer to the moorings and/or the tug than would otherwise have been necessary.

2.5 REACTION TO THE TURN

2.5.1 *Millennium Time*

As *Millennium Time* suddenly and unexpectedly turned towards *Redoubt*, the mate's perception was that the tug was turning towards the passenger vessel. As a result, he delayed taking any action for several seconds. Given that the interval between the start of *Millennium Time*'s turn to port and the collision was only 14 seconds, this delay was very significant.

As *Millennium Time* turned towards *Redoubt*, the master was facing the passengers on the top deck while providing a commentary. He was aware that *Redoubt* would be passing down the port side but he did not realise that *Millennium Time* was turning towards the tug until just before the vessels collided. Therefore, the commentary clearly distracted the master from his supervision of the mate. If the master had been at the helm, as required by company instructions, or even in the wheelhouse, the probability of *Millennium Time*'s unintended turn being detected in time for effective avoiding action to be taken would have been increased.

During the manoeuvring trials *Millennium Time* responded well to the helm and the interaction-induced sheer in September was easily stopped by the immediate use of counter-helm. Therefore, had the mate intervened sooner, the collision could probably have been prevented. By the time the mate took any action, the vessels were too close for it to be effective.

Although 69 years old, the mate was later proven to be medically fit and there is no evidence to indicate that he was distracted or fatigued. Therefore, his misinterpretation of the relative movement of *Millennium Time* and *Redoubt* was most likely influenced by his knowledge that he had not consciously initiated the turn using the steering wheel. In effect, *Millennium Time*'s turn did not fit with the mate's mental picture of events.

It is also likely that the wheelhouse ergonomics contributed to some degree to the mate's poor situational awareness. There was no rudder angle indicator so he could not easily check whether or not the rudder was amidships. Further, *Millennium Time*'s wheelhouse was positioned forward and had no visual references in the foreground, other than the top of the flag staff that was very close to and below the wheelhouse windows (**Figure 11**). Consequently it would have been harder for the mate to detect *Millennium Time* starting to swing if he was focusing on distant objects, such as the vessels ahead or number 4 arch of Blackfriars Road Bridge, at that time.

The magnetic compass, which would have indicated *Millennium Time*'s change in heading, was not working. However, as most navigation in this area is by 'line of sight' or 'by eye', it is recognised that a compass would rarely be used, if at all. This possibly explains why the compass had not been repaired or replaced and highlights that the defective compass probably made no difference to the mate's response.

2.5.2 Redoubt

Redoubt's master took action to avoid collision by putting the tug's helm hard to starboard as soon as he saw that *Millennium Time* was turning towards his vessel. Although this did not prevent the collision, it possibly prevented the vessels hitting head on, and therefore lessened the severity of the impact. The master's call to London VTS following the collision was also very prompt.

2.6 STEERING SYSTEM

In addition to the problems caused by the lack of any rudder angle indication in *Millennium Time*'s wheelhouse, it is evident from the problems associated with the rod and gear system itself that the system's performance was not sufficient for use on the congested and navigationally challenging waters of the Thames. The manoeuvring trials showed that *Millennium Time* responded quickly when more than 10° of rudder was applied. However, as the rudders were not balanced²³, the force needed to turn the steering wheel when underway at normal operating speed, the number of turns of the steering wheel required to move the rudders through their operating range, and the 'play' between the steering wheel and the rudder, all combined to reduce the system's speed of response and its ease of use.

The rod and gear system fitted to the M2 class vessels had many components and, due to its age, some spare parts were made to order. Given the complexity and overall length of the system with its numerous bevel gearboxes and universal joints, it was inevitable that the maintenance effort required increased over time. Notwithstanding the ongoing replacement of defective or worn parts, the system's performance would have deteriorated since its initial installation and is likely to continue to reduce further. The advantages of changing the rod and gear systems on board these vessels to powered steering was recognised by City Cruises in 2009. Although the twin-screw M2 vessels remained more manoeuvrable than many single-screw vessels operating on the River Thames, it is unfortunate that the plan to fit replacement hydraulic systems was not approved by the City Cruises' Board until 2014, and that none of the vessels had been modified prior to this accident occurring.

2.7 HELMSMAN CERTIFICATION

MSN 1808(M) and Thames byelaws required that the master of *Millennium Time* held a BML with a local knowledge endorsement. None of the vessel's other crew, including the mate, were required to hold a BML and there were no regulatory restrictions on them taking the helm. However, the operational memoranda issued by City Cruises in 2012, in response to accidents involving its vessels and recommendations from the PLA (Section 1.7.2), required a BML licence holder to be at the helm in busy parts of the river, and unqualified helmsmen to be properly supervised by the master in other areas. Although *Millennium Time*'s mate had extensive experience of the Thames and had worked on board the M2 class vessels for 14 years, he did not hold a BML and there is no record of him having completed City Cruises' internal boat-handling assessment. That the mate was at the helm at the time of *Millennium Time*'s collision while the master was providing a commentary to the passengers, shows that these company instructions were ignored.

²³ A 'balanced rudder' is a rudder on which the axis of rotation is behind its forward edge. This minimises the effort required to turn the rudder.

As three of *Millennium Time*'s four scheduled stops (**Annex A**) were between Cherry Garden Pier and Westminster Bridge, Operational Memoranda 11/2012 and 12/2012 had the effect of requiring the master to be at the helm for much of his 9 hour shift. The master would also have had to remain in the wheelhouse for most of the remaining time in order to properly supervise the mate when the vessel was between Cherry Garden Pier and Greenwich Pier. Although not impossible, such a routine would have been difficult to follow. Moreover, the potential additional disadvantage of lost revenue from the sightseeing commentaries made the instructions contained in the operational memoranda, although well-intended, unlikely to be followed.

The traffic density, reduced passing distances, bridges, moorings, strong tidal streams and the likelihood of interaction on the Thames, make it a very challenging area in which to navigate. The policy adopted by some operators to ensure that crew, other than a vessel's master, hold a BML, appears to be a sensible measure to help meet these challenges. Indeed, PLA's recommendation to the MCA in 2012 indicates that the port authority would welcome a regulatory requirement in this respect. However, it would be impractical to require all helmsmen on the Thames to hold a BML, otherwise there would be little scope for training newcomers.

Nonetheless, the provision of more than one BML holder on each vessel has several benefits that lend support to the recommendation in the PLA's investigation report (**Annex C**) for the authority to:

Amend Port of London Authority General Direction 18 – Persons on the bridge to require either the Master, or a deck officer, fully qualified and competent to take navigational conduct of the vessel, to be at the conning position at all times. [sic]

A vessel's master is at all times responsible for the safe navigation of his vessel. Therefore, until the PLA's recommendation is implemented, it is inappropriate for masters, who are the sole BML holders on board, to routinely engage in entertaining passengers while their vessels are transiting congested waters where high speeds of encounter with other vessels and fixed objects exist.

2.8 CREW RESPONSE

When *Millennium Time*'s master saw that the collision with *Redoubt* was imminent, his shouting of 'stand fast' and then diving to the deck contributed little to the safety of the passengers. In contrast, his securing of the vessel on the Coin Street moorings within 3 minutes of the collision was quick and positive, and facilitated their safe evacuation. However, although the vessel had not listed and no bilge alarms had activated, the master should have ensured that a physical check of the vessel's watertight integrity, as prompted in the collision checklist (**Annex B**), was carried out as soon as possible.

The RNLI and the MPU arrived on board *Millennium Time* within 7 minutes of the collision and almost as soon as the vessel had been moored. Until then, the master and mate were busy manoeuvring and securing the vessel. During this period, some of the 362 passengers were undoubtedly shocked and panicked by what had happened.

Millennium Time's crew of five met the required safe manning level indicated on the vessel's Passenger Vessel Safety Certificate to deal with up to 495 passengers. In addition, the in-house training provided to the customer service agents should also have provided a basic level of safety awareness and customer care. However, it is evident that the crew struggled to manage such a large number of linguistically diverse passengers, including schoolchildren, in what was inevitably a chaotic situation.

In such circumstances, strong leadership was required. Although the customer service agents were trained in the onboard procedures and the location of safety equipment, they were possibly also in a state of shock to some degree. They were also not necessarily equipped with the people skills often required during an emergency. This was acknowledged in the customer service agents' terms of reference (paragraph 1.7.3) which required them to follow the master's instructions. However, *Millennium Time*'s master, although having completed crowd management training, was focused on other matters and his ability to direct the customer service agents was limited.

Masters are almost certain to be pre-occupied and have a high workload during an emergency. Therefore, it is important in the short-term that the remaining crew are able to deal effectively with large numbers of potentially panicked and distressed passengers by themselves. Notwithstanding the training requirements of permanent crew detailed in MGN 203 (M), this will frequently be difficult to achieve given the seasonal nature of the customer service agents' employment. Nonetheless, it is important that customer service agents are properly trained to deal effectively with large numbers of passengers in an emergency situation when the deck officers are attending to other aspects of the emergency. To this end, the provision of crowd management training by City Cruises to its crews, in addition to masters, merits consideration.

2.9 PASSENGER COUNT

The Thames Safety Inquiry highlighted the importance of knowing the number of passengers on board a passenger vessel in the event of an emergency. Therefore, it is of concern that 25 years later the number of passengers reported to be on board *Millennium Time* when this collision occurred was so inaccurate.

The number of passengers counted by the MPU on board *Millennium Time* following the collision with *Redoubt* was 362. This was 64 people fewer than the number transmitted on the vessel's AIS, which indicated that 426 passengers were on board. It is evident that this discrepancy was due to the AIS having not been updated following the disembarkation and embarkation of passengers at Westminster and Waterloo piers. Consequently, the vessel did not meet the *Merchant Shipping (Counting and Registration of Persons on Board Passenger Ships) Regulations 1999*, which requires that an accurate record is held on board and ashore at all times.

Even had the passenger numbers been updated on *Millennium Time* following each of its stops, it is unlikely that the passenger count would have been correct. In common with many operators on the Thames, City Cruises relied on its crew to use a hand-operated tally counter to count the passengers on board its vessels. Counting thousands of passengers on and off over a dozen stops each day using this method is extremely difficult and is prone to error. Notwithstanding the difficulty

for crews to make sure that passengers are not missed or are double counted, the use of the tally counter is repetitive and monotonous, which potentially could lead to crews becoming inattentive and/or easily distracted from the task.

2.10 SAFETY MANAGEMENT

The circumstances of this accident and other accidents involving other M2 vessels in the Kings Reach of the Thames since 2012 (paragraph 1.7.2 and 1.15) indicates that City Cruises' safety management of its vessels, and the safety culture of its crews, have yet to develop fully. In particular:

- The disadvantages of a lack of rudder angle indication on board the M2 class vessels were not identified.
- The manning of *Millennium Time*'s wheelhouse at the time of the collision was at variance with company instructions.
- The practicability of complying with Operational Memoranda 11/2012 and 12/2012 relating to wheelhouse manning on its crews' work patterns was not fully considered.
- The passenger count was inaccurate.
- Physical checks for water ingress were not conducted immediately following the collision.
- The customer service agents did not manage the passengers effectively or prepare them for disembarkation.
- Records of the customer service agents' 'competent crew' training were not held.
- There was no record of the mate completing an internal boat-handling assessment.
- The performance of the steering system in the M2 class vessels was not sufficient to fully meet the challenges encountered on the busy waters of the Thames, but the systems had not been replaced.
- *Millennium Time*'s magnetic compass was not working.

Although City Cruises' SMS was compliant with the requirements of the DSMC, more effort is required to ensure that the system is used, adhered to and updated in order to minimise the risks to its vessels and the thousands of passengers it carries each day.

2.11 CONSTRUCTION STANDARDS

Although *Millennium Time* met the construction standards set out in MSN 1699, which were in force when the vessel was built in 1999, rather than the updated standards issued in 2010 in MSN 1823, this does not mean that the vessel was not safe. Indeed, the damage sustained by *Millennium Time* after its collision with *Redoubt* at a closing speed of about 16 knots was limited by its design and

construction. Furthermore, even if the hull's watertight integrity had been breached, the vessel's likelihood of survival would have been increased by its watertight subdivisions.

However, not all passenger vessels operating on the Thames meet the construction standards contained in either MSN 1699 or MSN 1823. Some passenger vessels, due to their age, pre-date these requirements and are not constructed with watertight sub-divisions. Consequently, a similar accident involving one of these older vessels has the potential to result in far more serious consequences than those experienced on this occasion. The possibility of an accident with significant loss of life, similar to *Marchioness*, remains. If Lord Justice Clarke's intent that 'grandfather rights' should not be allowed to compromise safety is to be achieved, the MCA's intention to curb these rights in a risk-based approach focusing on safety-critical areas, which it was unable to do in 2010, must be fully supported.

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. *Millennium Time's* sheer to port, which led to the collision with *Redoubt*, resulted from several degrees of port rudder being set without the mate being aware, and/or the effects of hydrodynamic interaction. [2.2]
2. *Millennium Time* was not required to have rudder angle indication in the wheelhouse but its absence meant that the mate had no way of knowing the angle of the rudders. [2.3]
3. The benefits of rudder indication in *Millennium Time's* wheelhouse during the trials in September 2014, together with the circumstances of this accident, support the case for the retro-fitting of rudder angle indication in the wheelhouses of other Class V passenger vessels equipped with non-powered steering. [2.3]
4. The risk of hydrodynamic interaction was increased in the vicinity of the Coin Street moorings in the King's Reach where the navigable water available was reduced by *Redoubt* being 15m to the south of its usual track. [2.4]
5. *Millennium Time's* mate's perception that the tug was turning towards the passenger vessel was influenced by his knowledge that he had not consciously initiated the turn using the steering wheel and the ergonomics of the wheelhouse. [2.5.1]
6. *Millennium Time's* master did not effectively supervise the mate on the helm because he was providing a sightseeing commentary to the passengers. [2.5.1]
7. The problems associated with the rod and gear steering system fitted on board *Millennium Time* reduced its effectiveness in the congested and navigationally challenging waters of the Thames. [2.6]
8. The advantages of changing the rod and gear steering systems on board the M2 class vessels was recognised by City Cruises in 2009 but none of the vessels had been modified at the time of this accident. [2.6]
9. As the mate on board *Millennium Time* did not hold a BML, company instructions required the master to be at the helm. These instructions were ignored. [2.7]
10. Operational memoranda regarding the wheelhouse manning on board *Millennium Time* were unlikely to be followed by the master and the mate due to the vessel's schedule and the potential loss of earnings from sightseeing commentaries. [2.7]
11. It was inappropriate for *Millennium Time's* master to be engaged entertaining the passengers, leaving the unqualified mate at the helm. [2.7]
12. The traffic density, reduced passing distances, bridges, moorings, strong tidal streams and the likelihood of interaction on the Thames make it a very challenging area in which to navigate. The policy adopted by some operators to ensure that crew, other than a vessel's master, hold a BML, appears to be a sensible measure to help meet these challenges. [2.7]

13. It is evident from the circumstances of this accident and the accidents involving M2 vessels since 2012 that City Cruises' safety management and the safety culture of its crews have yet to develop fully. [2.10]

3.2 OTHER SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT

1. It is possible that *Millennium Time*'s sudden sheer towards *Redoubt* was influenced by the effects of hydrodynamic interaction. However, the hydrodynamic forces at work were complex and it is difficult to accurately quantify the collective affects of their potential sources. [2.4]

3.3 SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. The failure to physically check the watertight integrity of *Millennium Time* following the collision was a significant omission, and the apparent ineffectiveness of the vessel's crew in managing the passengers is of concern. [2.8]
2. *Millennium Time*'s customer service agents were trained in emergency procedures and the location of safety equipment. However, this training did not fully equip them with the people skills that are often required in chaotic situations during an emergency. [2.8]
3. *Millennium Time* had 64 fewer passengers on board than the number transmitted on the vessel's AIS. The AIS data had not been updated following the disembarkation and embarkation of passengers at Westminster and Waterloo piers. [2.9]
4. Notwithstanding the difficulty for crews to make sure that passengers are not missed or are double counted, the counting of passengers with a hand tally counter is repetitive and monotonous and can potentially lead to crews becoming inattentive and/or easily distracted. [2.9]
5. Not all passenger vessels operating on the Thames are constructed with watertight sub-divisions. Consequently, the potential for a far more serious accident remains and the MCA's intentions to restrict 'grandfather rights' in safety-critical areas should be supported. [2.11]

SECTION 4 - ACTION TAKEN

4.1 MAIB ACTIONS

The **Marine Accident Investigation Branch** has:

Informed the Maritime and Coastguard Agency of the initial findings of manoeuvring and steering trials conducted off Gravesend on board *Millennium Time* on 19 August 2014. The Maritime and Coastguard Agency was recommended to:

2014/136 Require City Cruises to demonstrate that suitable and sufficient control measures are in place to ensure the safe navigation of the M2 river liners, taking into account, inter alia:

- the difficulty in steering the vessels
- the ergonomics of the wheelhouse
- the numbers of passengers carried
- the traffic density and proximity of navigational dangers on the Thames.

Following its investigation into the City Cruises vessel *Millennium Diamond* making contact with Tower Bridge on 4 June 2014, made a recommendation to the Maritime and Coastguard Agency to:

2015/116 Include in the Boatmasters' Licence generic syllabus a requirement for candidates to demonstrate a knowledge of the risks of distraction and the speed with which unsafe conditions can develop on inland waterways and in harbours.

4.2 ACTIONS TAKEN BY OTHER ORGANISATIONS

City Cruises Plc has:

Fitted the M2 river liners with rudder indicators and replaced the vessels' rod and gear steering with hydraulic steering systems.

SECTION 5 - RECOMMENDATIONS

The **Maritime and Coastguard Agency** is recommended to:

- 2015/131 Assess the steering arrangements on board domestic passenger vessels with non-powered steering and, where deemed to be beneficial and pragmatic, require these vessels to have rudder angle indication in the wheelhouse.

The **Maritime and Coastguard Agency** is also recommended to:

- 2015/132 Prioritise and resource its proposed legislation designed to limit the application of superseded regulations to existing domestic passenger vessels in safety-critical areas so that the legislation is introduced as quickly as possible.

The **Maritime and Coastguard Agency, Port of London Authority, Transport for London** and the **Passenger Boat Association** are recommended to:

- M2015/133 Work together to explore the use of technology to improve the accuracy of the passenger count on board passenger vessels on the River Thames.

The **Port of London Authority** is recommended to:

- 2015/134 Monitor traffic flow and density in the King's Reach and canvas river users in order to fully assess the impact of the Coin Street moorings on the navigable water available to vessels operating in the area.

City Cruises Plc is recommended to:

- 2015/135 Take action to improve the safe operation of its vessels by addressing the recommendations made by the Port of London Authority in its investigation report, but also focusing on:
- The advantages of having additional boatmasters' licence holders on board.
 - The problems inherent in masters giving commentaries while their vessel is underway.
 - Compliance with company instructions.
 - Wheelhouse ergonomics.
 - The importance of accurate passenger numbers.
 - Crew training records.
 - The management of passengers in an emergency.

City Cruises' spring/summer timetable 2014

T I M E T A B L E

From 31st of March until 2nd November 2014 inclusive

Departures from Westminster Pier

Westminster Pier to Tower Pier

0915*	0945	1015	1045	1115	1145	1215	1245	1320	1350	1420
1450	1520	1550	1620	1650	1720^	1750^	1815^	1850^	1920*†	1950*†

Westminster Pier to Greenwich Pier

0915*	0945	1015	1045	1115	1145	1215	1245	1320	1350	1420
1450	1520	1550	1620*	1650*						

Departures from London Eye Pier

London Eye Pier to Tower Pier

0925*	0955	1030	1100	1130	1200	1230	1300	1335	1405	1435
1505	1535	1605	1635	1705	1735^	1805^	1830^	1905^	1935*†	2010*†

London Eye Pier to Greenwich Pier

0925*	0955	1030	1100	1130	1200	1230	1300	1335	1405	1435
1505	1535	1605	1635*	1705*						

Departures from Tower Pier

Tower Pier to Westminster Pier & London Eye Pier

1005	1035	1105	1135	1205	1240	1310	1340	1410	1440	1510
1540	1610	1640	1710	1740	1810	1840*	1910*	(1940*† to Westminster Only)		

Tower Pier to Greenwich Pier

0955*	1025	1100	1130	1200	1230	1300	1330	1405	1435	1505
1535	1605	1635	1705*	1740*						

Departures from Greenwich Pier

Greenwich to Tower Pier

1025	1055	1125	1200	1230	1300	1330	1400	1430	1500	1530
1600^	1630^	1700^	1730^	1800*†	1830*†					

Greenwich Pier to Westminster Pier & London Eye Pier

1025	1055	1125	1200	1230	1300	1330	1400	1430	1500^	1530^
1600^	1630^	1700^	1730^	1800*†	1830*†					

Thames Circular Cruise

Operates daily departing from Tower Pier

1000	1100	1200	1300	1500	1600
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Returns to Tower Pier after 45 minutes.

- * operates only from 21st July to 7th September 2014 inclusive.
- † no return service available from 21st July to 7th September 2014 inclusive.
- ^ no return service available from 31st March to 20th July 2014 inclusive and also from 8th September to 2nd November 2014 inclusive.

IMPORTANT

For the last boats of the day, please check return times with a member of staff. Actual times may vary depending upon the state of the tide and other river traffic. Always check with City Cruises staff for any changes to advertised service.

Collision checklist

Operations - 02077400487
07894899742

London VTS – VHF Channel 14
02088550315

Collision

Operations to be called at earliest convenience - 07894899742

1. Sound the general emergency alarm
2. Check for missing or injured personnel.
 - **No** injured or missing passengers go to **number 3**.
 - **Missing person** - Make an announcement to passengers, all crew look in all void spaces for missing person
 - If not found go to **Man overboard procedure**, go to page 9
 - **If injured offer first aid**
3. Close watertight door and fire doors
4. If possible Manoeuvre Vessel to safe Haven
 - **If alongside a pier** – Disembark passengers to transport or other vessel
 - **If on the River** – Arrange for passenger transfer to another vessel
5. Drop the Anchor
6. Display relevant lights or day marks
7. Contact VTS - Switch VHF to Channel 14 or Call 0208 8550 315
8. Call Operations – 02077400487, 07894899742
9. Check bilges and tanks
10. Check for Fire / Damage
 - If there is a Fire, go to **Fire Emergency Action** on page 5
11. Secure vessel whilst pending recovery/ repair
12. Submit full report to Operations.

PLA Investigation Report Recommendations

RECOMMENDATIONS

City Cruises is recommended to:

1. Provide focused training on bridge team management to the Mate and Master to ensure they work together as a team and are not distracted from their primary task of navigating the vessel. *[Conclusion 1]*
2. Undertake urgent modifications to the M2 Riverliner steering system, to the satisfaction of the MCA, to provide a steering system which allows crews to safely navigate the vessel without undue exertion or 'kick back' through the steering system, and allows the vessel to respond quickly to any urgent situations. *[Conclusion 2]*
3. Undertake an assessment of all crews to ensure they are sufficiently fit and able to handle the steering system of the M2 Riverliner, until such time as the steering systems are upgraded. *[Conclusion 2]*
4. Ensure that the Master, or a deck officer, fully qualified and competent to take navigational conduct of the vessel, is at all times at the conning position of all City Cruises vessels *[Conclusion 3]*
5. Provide a clear policy for the provision of passenger commentaries aboard City Cruises vessels, to ensure the safety of navigation of City Cruises vessels are not compromised by the commentary, and the requirements contained in recommendation 3 of this report are met. *[Conclusion 4]*
6. Identify options for improving the 'crisis management' skills of City Cruises crews; including 'cabin staff' to ensure they are able to respond effectively to emergency situations and take appropriate actions to ensure the safety of the vessel and passengers. This could be achieved with more effective drills and training. *[Conclusion 5]*
7. Implement a requirement for all members of City Cruises bridge crew to hold a valid ML5 medical, to meet the requirements of the National Boatmaster's Licence regulations, and ensure all staff are medically fit and able to undertake their duties. *[Conclusion 6]*
8. Ensure all City Cruises crews provide accurate person on board data, in compliance with the Thames Byelaw 12, at all times and effectively manage this process. *[Conclusion 8]*
9. Consider the implementation of a wheelhouse CCTV system with audio capability in all City Cruises vessels to aid in determining the root cause of future incidents. *[Conclusion 9]*
10. Amend their procedures, to ensure changeovers of the helm of the vessel are normally conducted at Greenwich, to allow helms to re-familiarise with the vessel before navigating the more challenging waters of central London. *[Conclusion 11]*



11. Install rudder indicators at the steering position, when undertaking modifications to the steering system. *[Conclusion 12]*

12. Raise awareness of this incident with City Cruises crews to ensure the lessons identified have been fully promulgated to all crews.

Maritime & Coastguard Agency is recommended to:

1. Consider a thorough review of the steering systems of the M2 Riverliners to determine whether the steering system is acceptable given the confined and challenging conditions on the tidal Thames. *[Conclusion 2]*
2. Consider a review of the current standards for steering systems and manoeuvrability of Class V passenger vessels in both MGN 1823 and existing vessels navigating on the tidal Thames, to ensure they remain 'Fit for Purpose' and take account of the inherent conditions on the tidal Thames. *[Conclusion 2]*

Cory Riverside is recommended to:

1. Amend their passage plan to highlight the need to provide clear water for outbound traffic adjacent to the Coin St Moorings in Kings Reach. *[Conclusion 7]*
2. Raise awareness of this incident with Cory Riverside tug crews to ensure the lessons identified have been fully promulgated to all crews. *[Conclusion 7]*

Port of London Authority is recommended to:

1. Amend Port of London Authority General Direction 18 – Persons on the Bridge to require either the Master, or a deck officer, fully qualified and competent to take navigational conduct of the vessel, to be at the conning position at all times. *[Conclusion 3]*
2. Consider modifications to the Thames AIS specification, to include the provision of an 'auto-prompt' feature, to remind crews to input up to date person on board data on each pier departure. *[Conclusion 8]*
3. Review the central London Reporting Points and Special Signal Light activation ranges, during NRAWG 53 – Large vessel transits of Central London Bridges, to determine if any changes should be made to either system. *[Conclusion 9]*
4. Develop and distribute a Safety Bulletin to raise awareness of the lessons identified in this report
5. Undertake a detailed review of Hazard 4 – Collision - Class V & VI Passenger Vessel, Hazard 183 – Collision - Putney Bridge to London Bridge and Hazard 21 – Collision – Tugs and Barges (River)
6. Provide a copy of this report to City Cruises and MCA drawing their attention to the recommendations raised
7. Provide a copy of this report to the MAIB



